

COLLEGE OF SCIENCE & ENGINEERING

INDIVIDUAL TASK COVER SHEET

Student									
Please sign, date and attach cover sheet to front of written assessment task OR submit as a separate document for non-written assessment task. A cover sheet is to be completed for <u>each</u> assessment task.									
SUBJECT CODE	CP5405								
SUBJECT TITLE	Scalable Database Systems								
STUDENT FAMILY NAME	Student Given Name	JCU Student Number							
Zifei	Yu	1	4	1	4	8	8	1	7
ASSESSMENT TITLE	Assessment 2-Software Development/Creation								
DUE DATE	29/03/2026								
LECTURER NAME	Mr. Insu Song								
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Part A – Data Ingestion & Schema Design

1. Import the provided CSV data into MongoDB.

My MongoDB Collab: [MongoDB indexing and data analysis strategies for Location and Ticket data with Comment for Performance improvements](#)

This section implements and validates the initial ingestion workflow for the DataFest event platform in MongoDB. The main goal of the code is to read structured CSV data, convert important fields into the correct MongoDB-compatible formats, and insert the cleaned records into the event_platform database. In particular, the script demonstrates how raw ticket scan records can be ingested step by step into the ticket_scans collection while preserving data quality, preventing accidental duplication during testing, and showing operational control over the import process. The assessment brief specifically requires the use of insertOne, insertMany, writeConcern, deleteOne, deleteMany, and drop during CSV ingestion, and these operations were all implemented and demonstrated in this section.

The code begins by reading CSV rows using Python's csv.DictReader, then converts timestamp values into datetime objects and rating values into integers where relevant. This preprocessing step is important because MongoDB queries, sorting, and later aggregation pipelines require correct data types rather than raw text values. Before inserting records, the script prints sample rows to confirm that parsing is correct and to reduce the risk of importing malformed or duplicate data. This step-by-step validation directly follows the task instruction to print and check records before ingestion.

After validation, the script connects to the ticket_scans collection with writeConcern(w="majority"), which improves reliability by ensuring writes are acknowledged at the replica-set majority level. The collection is then dropped before reloading test data so that repeated execution does not create duplicated records. The ingestion logic demonstrates both insert_one() and insert_many() operations: the first parsed row is inserted individually to prove single-document insertion, while the remaining rows are inserted in bulk with ordered=False so that one failed record would not stop the rest of the batch. This is more realistic for event-stream ingestion, where resilience and throughput are important.

The script also extends the starter ingestion workflow with additional CRUD operations required for testing and operational verification. update_one() is used to mark a specific imported ticket as reviewed, while update_many() enriches all records from gate G1 with additional fields such as gate_label and analytics_ready. Temporary records are then inserted and removed using delete_one() and delete_many() to prove that unwanted or test-only data can be safely cleaned up after ingestion. In addition, bulk_write() is used to execute mixed operations in one request, including insert, update, and delete actions. This shows that the ingestion workflow was not only completed, but also extended beyond the starter script to demonstrate broader MongoDB operational control.

Overall, this section successfully completed the required ingestion and operational tasks for the `ticket_scans` dataset. The code shows that the CSV data was parsed, validated, inserted into MongoDB, and managed through multiple write and cleanup operations. The final output confirms that the collection remained consistent after temporary test documents were removed, and the resulting records were enriched with additional metadata fields such as `schema_version`, `source_file`, and `import_batch`, making them more suitable for downstream analytics and schema evolution. This provides a strong foundation for the later schema design, indexing, and aggregation stages of the project.

[Put your python code here]

Demonstrate the use of `insertOne`, `InsertMany`, `writeConcern`, `deleteOne`, `deleteMany`, `drop` while reading and ingesting the CSV files.

Fig1: Installing PyMongoDB

```
# Installing PyMongo, this is the interface to connect to MongoDB with Python

#! python -m pip install pymongo==3.7.2
#! python -m pip install "pymongo[srv]"
! python -m pip install "pymongo[srv]=3.12"
```

Fig2: ingest python code to load the csv files from google drive

```
# Mount Google drive to read from your Google drive.
import csv

#for google drive mount
from google.colab import drive
drive.mount('/content/drive')

Mounted at /content/drive

# Set path information to the data files
path_to_data = "/content/drive/MyDrive/Colab Notebooks/CP5405 Assessment 2/A2Data/"

path_to_data1 = path_to_data + "ticket_scans.csv"
path_to_data2 = path_to_data + "rfid_movements.csv"
path_to_data3 = path_to_data + "feedback.csv"
```

Fig3: Import 'MongoClient' connection string in with Atlas credentials.

```
import csv
from datetime import datetime
#db = client.event_platform

def ingest_csv_to_collection(file_path, collection_name):
    #collection = db[collection_name]
    with open(file_path, 'r') as csvfile:
        reader = csv.DictReader(csvfile)
        for row in reader:
            # Convert timestamp string to datetime if exists
            if 'timestamp' in row:
                row['timestamp'] = datetime.fromisoformat(row['timestamp'])
            if 'rating' in row:
                row['rating'] = int(row['rating'])
            #collection.insert_one(row)
            print(row) # check if the data is correct before insert!!
            print(f"Ingested {file_path} into {collection_name}")

    ingest_csv_to_collection('ticket_scans.csv', 'ticket_scans')
    #ingest_csv_to_collection('rfid_movements.csv', 'rfid_movements')
    #ingest_csv_to_collection('feedback.csv', 'feedback')
```

The primary goal of the following code is to successfully parse, clean, and ingest the raw CSV event data (**ticket_scans**) into the MongoDB **event_platform** database. During the ingestion pipeline, data types are strictly cast (e.g., converting ISO strings to native MongoDB datetime objects) to ensure accurate time-based querying later. Furthermore, the script enriches the raw data with schema versioning and batch metadata (**schema_version**, **source_file**, **import_batch**) to ensure data traceability.

In addition to data ingestion, this code acts as a comprehensive demonstration of MongoDB's core CRUD capabilities. It explicitly executes and validates:

- Insertions: **insert_one** and **insert_many**
- Updates: **update_one** and **update_many**
- Deletions: **delete_one** and **delete_many**
- Advanced Batching: **bulk_write** combining mixed operations.

To ensure production-level reliability, **WriteConcern(w="majority")** is applied to guarantee data durability across replica sets, and **ordered=False** is utilized during bulk operations to provide fault tolerance (ensuring that a single document failure does not halt the entire batch ingestion).

```
# Part A: Section 1: Import the provided CSV data into MongoDB.
#use Insertone, insertMany, writeconcern, ordered...
#use deleteOne, deletMany, writeconcern, ordered...
#use updateOne, updateMany,
import csv
from datetime import datetime
db = client.event_platform

def ingest_csv_to_collection(file_path, collection_name):
    #collection = db[collection_name]
    with open(file_path, 'r') as csvfile:
        reader = csv.DictReader(csvfile)
        for row in reader:
            # Convert timestamp string to datetime if exists
            if 'timestamp' in row:
                row['timestamp'] =
datetime.fromisoformat(row['timestamp'])
            if 'rating' in row:
                row['rating'] = int(row['rating'])
            #collection.insert_one(row)
            print(row) # check if the data is correct before insert!!
            print(f"Ingested {file_path} into {collection_name}")

ingest_csv_to_collection(path_to_data1, 'ticket_scans')
#ingest_csv_to_collection('rfid_movements.csv', 'rfid_movements')
#ingest_csv_to_collection('feedback.csv', 'feedback')

print("Demonstrating MongoDB Operations")
```

```

# demonstrate as many Mongodb operations as possible: E.g., delete,
drop, update, .... while importing
# INSERT_ONE demo
from pymongo.write_concern import WriteConcern
from pymongo import InsertOne, UpdateOne, UpdateMany, DeleteOne
from pymongo.errors import BulkWriteError
from pprint import pprint

# define collection with writeConcern
ticket_scans_col = db.get_collection(
    "ticket_scans",
    write_concern=WriteConcern(w="majority")
)

# optional: clear old data first
ticket_scans_col.drop()
print("Old ticket_scans collection dropped.")

# recreate handle after drop
ticket_scans_col = db.get_collection(
    "ticket_scans",
    write_concern=WriteConcern(w="majority")
)

# read the CSV again and store rows for actual MongoDB operations
ticket_rows = []

with open(path_to_data1, 'r', encoding='utf-8') as csvfile:
    reader = csv.DictReader(csvfile)
    for row in reader:
        if 'timestamp' in row and row['timestamp']:
            row['timestamp'] = datetime.fromisoformat(row['timestamp'])
        if 'rating' in row and row['rating']:
            row['rating'] = int(row['rating'])

        row['schema_version'] = 1
        row['source_file'] = 'ticket_scans.csv'
        row['import_batch'] = 'batch_1'

        ticket_rows.append(row)

print("\nPreview ticket_rows:")

```

```

for doc in ticket_rows[:3]:
    pprint(doc)
if ticket_rows:
    result_one = ticket_scans_col.insert_one(ticket_rows[0])
    print("\ninsert_one demo:")
    print("Inserted document _id:", result_one.inserted_id)

# INSERT_MANY demo
# ordered=False means if one row fails, others can still continue
if len(ticket_rows) > 1:
    result_many = ticket_scans_col.insert_many(ticket_rows[1:],
ordered=False)
    print("\ninsert_many demo:")
    print("Inserted", len(result_many.inserted_ids), "documents")

# UPDATE_ONE demo
# Example: mark one imported record as checked
update_one_result = ticket_scans_col.update_one(
    {"ticket_id": "T0001"},
    {"$set": {"review_status": "checked"}}
)

print("\nupdate_one demo:")
print("matched:", update_one_result.matched_count)
print("modified:", update_one_result.modified_count)

# UPDATE_MANY demo
# Example: enrich all G1 gate records
update_many_result = ticket_scans_col.update_many(
    {"gate_id": "G1"},
    {"$set": {"gate_label": "Main Gate", "analytics_ready": True}}
)

print("\nupdate_many demo:")
print("matched:", update_many_result.matched_count)
print("modified:", update_many_result.modified_count)

# DELETE_ONE demo
# Insert a temporary document first, then delete it
temp_doc = {
    "ticket_id": "TEMP001",
    "attendee_id": "ATEST1",
    "gate_id": "G9",

```

```

        "timestamp": datetime.now(),
        "schema_version": 1,
        "source_file": "manual_temp",
        "import_batch": "temp_delete_one"
    }

ticket_scans_col.insert_one(temp_doc)

delete_one_result = ticket_scans_col.delete_one({"ticket_id":
"TEMP001"})

print("\ndelete_one demo:")
print("deleted:", delete_one_result.deleted_count)

# DELETE_MANY demo
# Insert temporary docs first, then delete them together
temp_many_docs = [
    {
        "ticket_id": f"TEMP_MANY_{i}",
        "attendee_id": f"ATEMP{i}",
        "gate_id": "G9",
        "timestamp": datetime.now(),
        "schema_version": 1,
        "source_file": "manual_temp",
        "import_batch": "temp_delete_many"
    }
    for i in range(3)
]

ticket_scans_col.insert_many(temp_many_docs, ordered=True)

delete_many_result = ticket_scans_col.delete_many(
    {"import_batch": "temp_delete_many"}
)

print("\ndelete_many demo:")
print("deleted:", delete_many_result.deleted_count)

# BULK_WRITE demo
# Show multiple operations in one call
# ordered=False allows later ops to continue if one fails
bulk_ops = [
    InsertOne({

```

```

        "ticket_id": "BULK001",
        "attendee_id": "ABULK1",
        "gate_id": "G2",
        "timestamp": datetime.now(),
        "schema_version": 1,
        "source_file": "bulk_demo",
        "import_batch": "bulk_batch"
    )),
    UpdateOne(
        {"ticket_id": "T0002"},
        {"$set": {"bulk_updated": True}}
    ),
    UpdateMany(
        {"gate_id": "G2"},
        {"$set": {"gate_category": "Secondary Gate"}}
    ),
    DeleteOne(
        {"ticket_id": "BULK001"}
    )
]

try:
    bulk_result = ticket_scans_col.bulk_write(bulk_ops, ordered=False)
    print("\nbulk_write demo:")
    print("inserted:", bulk_result.inserted_count)
    print("modified:", bulk_result.modified_count)
    print("deleted:", bulk_result.deleted_count)
except BulkWriteError as e:
    print("Bulk write error:")
    pprint(e.details)

# Final checks
print("\nFinal document count:")
print(ticket_scans_col.count_documents({}))

print("\nSample documents after import and operations:")
for doc in ticket_scans_col.find({}, {"_id": 0}).limit(5):
    pprint(doc)

```

[Screenshot of the results/outputs]

Fig4: Preview ticket_rows


```

Preview ticket_rows:
{'attendee_id': 'A170',
 'gate_id': 'G1',
 'import_batch': 'batch_1',
 'schema_version': 1,
 'source_file': 'ticket_scans.csv',
 'ticket_id': 'T0001',
 'timestamp': datetime.datetime(2025, 9, 1, 14, 16, 11)}
{'attendee_id': 'A323',
 'gate_id': 'G3',
 'import_batch': 'batch_1',
 'schema_version': 1,
 'source_file': 'ticket_scans.csv',
 'ticket_id': 'T0002',
 'timestamp': datetime.datetime(2025, 9, 1, 12, 10, 30)}
{'attendee_id': 'A149',
 'gate_id': 'G2',
 'import_batch': 'batch_1',
 'schema_version': 1,
 'source_file': 'ticket_scans.csv',
 'ticket_id': 'T0003',
 'timestamp': datetime.datetime(2025, 9, 1, 10, 22, 38)}

```

This screenshot shows sample records from **ticket_scans.csv** after being read into Python and transformed before insertion. The preview confirms that the ingestion script successfully converted the **timestamp** field into **datetime** format and appended additional metadata fields such as **schema_version**, **source_file**, and **import_batch**. This provides evidence that the raw CSV data was validated and prepared correctly before being written to MongoDB.

Fig5: insert/update/delete/bulk_write output

```

insert_one demo:
Inserted document _id: 69b5907fc3e447aa50bb31aa

insert_many demo:
Inserted 999 documents

update_one demo:
matched: 1
modified: 1

update_many demo:
matched: 244
modified: 244

delete_one demo:
deleted: 1

delete_many demo:
deleted: 3

bulk_write demo:
inserted: 1
modified: 243
deleted: 1

```

This screenshot demonstrates that the script successfully executed **insert_one**, **insert_many**, **update_one**, **update_many**, **delete_one**, **delete_many**, and **bulk_write**. These outputs provide evidence that the ingestion process was extended beyond basic loading and that the system can support realistic operational tasks such as data enrichment, temporary test-record cleanup, and batched write handling. This is important because the assessment explicitly requires these MongoDB operations to be demonstrated during ingestion.

Fig6: Final document count + sample documents

```

Final document count:
1000

```

```

Sample documents after import and operations:
{'analytics_ready': True,
 'attendee_id': 'A170',
 'gate_id': 'G1',
 'gate_label': 'Main Gate',
 'import_batch': 'batch_1',
 'review_status': 'checked',
 'schema_version': 1,
 'source_file': 'ticket_scans.csv',
 'ticket_id': 'T0001',
 'timestamp': datetime.datetime(2025, 9, 1, 14, 16, 11)}
{'attendee_id': 'A323',
 'bulk_updated': True,
 'gate_id': 'G3',
 'import_batch': 'batch_1',
 'schema_version': 1,
 'source_file': 'ticket_scans.csv',
 'ticket_id': 'T0002',
 'timestamp': datetime.datetime(2025, 9, 1, 12, 10, 30)}
{'attendee_id': 'A149',
 'gate_category': 'Secondary Gate',
 'gate_id': 'G2',
 'import_batch': 'batch_1',
 'schema_version': 1,
 'source_file': 'ticket_scans.csv',
 'ticket_id': 'T0003',
 'timestamp': datetime.datetime(2025, 9, 1, 10, 22, 38)}
{'analytics_ready': True,
 'attendee_id': 'A142',
 'gate_id': 'G1',
 'gate_label': 'Main Gate',
 'import_batch': 'batch_1',
 'schema_version': 1,
 'source_file': 'ticket_scans.csv',
 'ticket_id': 'T0004',
 'timestamp': datetime.datetime(2025, 9, 1, 8, 57, 46)}
{'attendee_id': 'A386',
 'gate_id': 'G4',
 'import_batch': 'batch_1',
 'schema_version': 1,
 'source_file': 'ticket_scans.csv',
 'ticket_id': 'T0005',
 'timestamp': datetime.datetime(2025, 9, 1, 10, 47, 34)}

```

This screenshot confirms that the final **ticket_scans** collection contains **1,000 valid** records after temporary test rows were deleted. The sample documents also show that update operations were applied successfully, including fields such as **review_status**, **gate_label**, **analytics_ready**, and **gate_category**. This provides final evidence that the ingestion workflow was completed correctly, that temporary validation records did not contaminate the dataset, and that the imported collection is ready for later schema optimisation and analytical processing.

2. Design an optimized schema using advanced data modeling (embedding, referencing, normalization/denormalization).

[Put your python code here]

Choice: Referenced raw event streams with a denormalized summary collection.

A referenced design is more appropriate for this system because **ticket_scans**, **rfid_movements**, and **feedback** are all high-cardinality event streams linked by **attendee_id**. Keeping these streams in separate collections preserves normalized raw data, avoids oversized attendee documents, and supports scalable ingestion. At the same time, **attendees_summary** serves as a denormalized analytics layer that stores precomputed

attendee-level indicators such as **ticket_scan_count**, **movement_count**, **distinct_location_count**, **avg_rating**, and **sentiment_label**. This hybrid design supports both operational storage and efficient analytical querying, while remaining flexible enough for future sources and schema evolution.

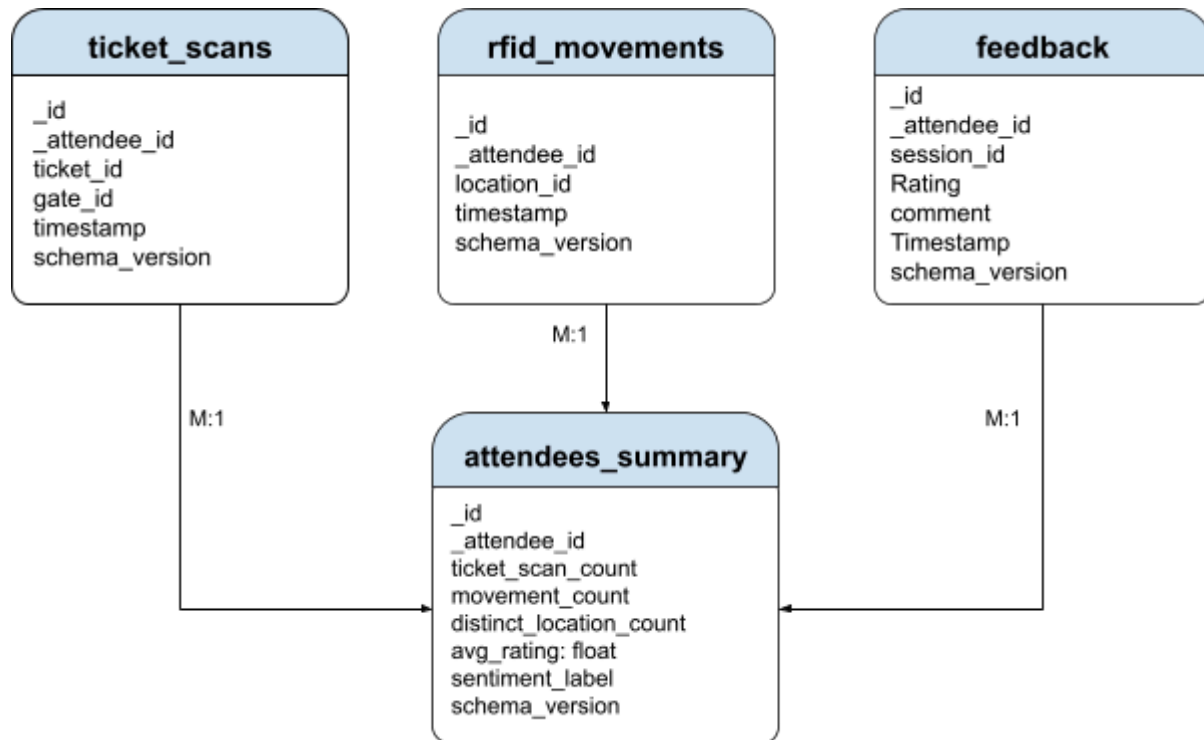


Fig 7: Hybrid Schema Architecture combining referenced raw collections and a denormalized summary collection.

As illustrated in Figure 7, We implemented a hybrid data architecture that effectively separates high-velocity write operations from read-heavy analytical queries. The figure demonstrates how the raw event streams (**ticket_scans**, **rfid_movements**, and **feedback**) are maintained as independent referenced collections to manage unbounded cardinality. Simultaneously, they are linked via **attendee_id** to the central **attendees_summary** collection, which serves as a pre-computed layer for immediate dashboard access.

This section designs and implements an optimized MongoDB schema for the DataFest event analytics system using advanced data modeling principles, specifically referencing, normalization, and denormalization. The main goal of the code is to keep the raw event streams in separate collections **ticket_scans**, **rfid_movements**, and **feedback**, while creating a new denormalized analytics collection called **attendees_summary** for efficient attendee-level reporting. This directly addresses the assessment requirement to design an optimized schema using advanced data modeling and to ensure that the design is robust enough for future scaling and unforeseen data sources.

The code begins by preparing and loading the raw **rfid_movements** and **feedback** datasets into their own collections, while preserving **ticket_scans** from Section 1 as the original ticket

event stream. It then creates a new summary collection, `attendees_summary`, and removes any old version of that collection before rebuilding it. This reset step is important because it ensures that repeated testing does not create duplicate summary documents and that the derived collection always reflects the latest raw data. The script then processes ticket scan records first, using `attendee_id` as the shared reference key, and initializes one summary document per attendee with denormalized fields such as `ticket_scan_count`, `gates_used`, `first_scan_time`, and `last_scan_time`.

Next, the code merges RFID movement data into the same attendee-level summary documents. A helper function is used to detect the most likely location-related field from incoming RFID records, which improves flexibility if future data sources use slightly different field names such as `location_id`, `zone_id`, or `reader_id`. This makes the schema more robust and adaptable, which is important because the assessment explicitly expects the design to support future and unforeseen scaling requirements. The transformation logic then computes denormalized movement metrics such as `movement_count`, `distinct_location_count`, `locations_visited`, `first_movement_time`, and `last_movement_time`. These are analytical fields that would otherwise require repeated aggregation over the raw event stream.

The code then merges feedback records into the same attendee-level profile by referencing the shared `attendee_id`. It calculates summary-level sentiment indicators such as `feedback_count`, `avg_rating`, `latest_feedback_time`, and `sentiment_label`. This step shows the use of denormalization for read optimization: instead of forcing every dashboard query to run expensive joins and aggregations across raw collections, the most commonly needed attendee-level indicators are materialised once inside `attendees_summary`. This aligns with the README justification in the template, which explains that referenced raw collections should be combined with a denormalized summary layer to improve analytical performance while preserving scalable event-level storage.

The design choice for this schema is therefore a referenced raw event-stream model with a denormalized summary collection. This is more appropriate than embedding all raw event data inside one attendee document because each attendee can generate many ticket scans, RFID movements, and feedback records over time. Storing all raw records inside a single document would increase document growth and reduce scalability, while referenced raw collections preserve event-level granularity and improve maintainability. At the same time, the summary collection acts as an analytics layer that supports fast reporting and downstream aggregation. Overall, successfully completes the schema design task by implementing a hybrid model that balances data integrity, analytical efficiency, maintainability, and future extensibility.

```
# Part A: Section 2: Design an optimized schema using advanced data
modeling
# (embedding, referencing, normalization/denormalization).
```

```

#client =
MongoClient("mongodb+srv://zifeiyu:Yuleyang098#@cluster0.dam4hay.mongod
b.net/?appName=Cluster0")
from pprint import pprint

db = client.event_platform

# Raw collections from Section 1
ticket_scans_col = db.ticket_scans
rfid_movements_col = db.rfid_movements
feedback_col = db.feedback

# New optimized summary collection
attendees_summary_col = db.attendees_summary

# Drop old summary collection to avoid duplicates during testing
attendees_summary_col.drop()
print("Old attendees_summary collection dropped.")

# Helper: detect likely location field in RFID docs
def get_location_value(doc):
    candidate_fields = ["location_id", "zone_id", "zone", "room_id",
"reader_id", "gate_id"]
    for field in candidate_fields:
        if field in doc and doc[field] is not None:
            return doc[field]
    return None

# Step 1: Build attendee summary from ticket scans
attendees_data = {}

for row in ticket_scans_col.find():
    aid = row.get("attendee_id")
    if not aid:
        continue

    if aid not in attendees_data:
        attendees_data[aid] = {
            "attendee_id": aid,
            "schema_version": 1,
            "summary_type": "attendee_profile",

            # denormalized summary fields

```

```

        "ticket_scan_count": 0,
        "gates_used": [],
        "first_scan_time": None,
        "last_scan_time": None,

        # placeholders for merged summaries
        "movement_count": 0,
        "distinct_location_count": 0,
        "locations_visited": [],
        "first_movement_time": None,
        "last_movement_time": None,

        "feedback_count": 0,
        "avg_rating": None,
        "sentiment_label": None,
        "latest_feedback_time": None
    }

    attendees_data[aid]["ticket_scan_count"] += 1

    gate_id = row.get("gate_id")
    if gate_id and gate_id not in attendees_data[aid]["gates_used"]:
        attendees_data[aid]["gates_used"].append(gate_id)

    ts = row.get("timestamp")
    if ts:
        if attendees_data[aid]["first_scan_time"] is None or ts <
attendees_data[aid]["first_scan_time"]:
            attendees_data[aid]["first_scan_time"] = ts
        if attendees_data[aid]["last_scan_time"] is None or ts >
attendees_data[aid]["last_scan_time"]:
            attendees_data[aid]["last_scan_time"] = ts

print(f"Processed ticket scan summaries for {len(attendees_data)}
attendees.")

# Step 2: Merge RFID movement summaries
for row in rfid_movements_col.find():
    aid = row.get("attendee_id")
    if not aid:
        continue

    if aid not in attendees_data:

```

```

attendees_data[aid] = {
    "attendee_id": aid,
    "schema_version": 1,
    "summary_type": "attendee_profile",

    "ticket_scan_count": 0,
    "gates_used": [],
    "first_scan_time": None,
    "last_scan_time": None,

    "movement_count": 0,
    "distinct_location_count": 0,
    "locations_visited": [],
    "first_movement_time": None,
    "last_movement_time": None,

    "feedback_count": 0,
    "avg_rating": None,
    "sentiment_label": None,
    "latest_feedback_time": None
}

attendees_data[aid]["movement_count"] += 1

loc = get_location_value(row)
if loc and loc not in attendees_data[aid]["locations_visited"]:
    attendees_data[aid]["locations_visited"].append(loc)

ts = row.get("timestamp")
if ts:
    if attendees_data[aid]["first_movement_time"] is None or ts <
attendees_data[aid]["first_movement_time"]:
        attendees_data[aid]["first_movement_time"] = ts
    if attendees_data[aid]["last_movement_time"] is None or ts >
attendees_data[aid]["last_movement_time"]:
        attendees_data[aid]["last_movement_time"] = ts

# finalize distinct count
for aid in attendees_data:
    attendees_data[aid]["distinct_location_count"] =
len(attendees_data[aid]["locations_visited"])

print("Merged RFID movement summaries.")

```

```

# Step 3: Merge feedback summaries
feedback_accumulator = {}

for row in feedback_col.find():
    aid = row.get("attendee_id")
    if not aid:
        continue

    if aid not in attendees_data:
        attendees_data[aid] = {
            "attendee_id": aid,
            "schema_version": 1,
            "summary_type": "attendee_profile",

            "ticket_scan_count": 0,
            "gates_used": [],
            "first_scan_time": None,
            "last_scan_time": None,

            "movement_count": 0,
            "distinct_location_count": 0,
            "locations_visited": [],
            "first_movement_time": None,
            "last_movement_time": None,

            "feedback_count": 0,
            "avg_rating": None,
            "sentiment_label": None,
            "latest_feedback_time": None
        }

    if aid not in feedback_accumulator:
        feedback_accumulator[aid] = {
            "count": 0,
            "rating_sum": 0,
            "latest_feedback_time": None
        }

    rating = row.get("rating")
    if rating is not None:
        feedback_accumulator[aid]["count"] += 1
        feedback_accumulator[aid]["rating_sum"] += rating

```



```

        ts = row.get("timestamp")
        if ts:
            if feedback_accumulator[aid]["latest_feedback_time"] is None or
ts > feedback_accumulator[aid]["latest_feedback_time"]:
                feedback_accumulator[aid]["latest_feedback_time"] = ts

for aid, stats in feedback_accumulator.items():
    attendees_data[aid]["feedback_count"] = stats["count"]

    if stats["count"] > 0:
        avg_rating = round(stats["rating_sum"] / stats["count"], 2)
        attendees_data[aid]["avg_rating"] = avg_rating

        if avg_rating >= 4:
            attendees_data[aid]["sentiment_label"] = "positive"
        elif avg_rating <= 2:
            attendees_data[aid]["sentiment_label"] = "negative"
        else:
            attendees_data[aid]["sentiment_label"] = "neutral"

    attendees_data[aid]["latest_feedback_time"] =
stats["latest_feedback_time"]

print("Merged feedback summaries.")

# Step 4: Insert optimized summary documents
attendees_summary_docs = list(attendees_data.values())

if attendees_summary_docs:
    attendees_summary_col.insert_many(attendees_summary_docs)
    print(f"Inserted {len(attendees_summary_docs)} documents into
attendees_summary.")

# Step 5: Show results
print("\nCollections in event_platform:")
print(db.list_collection_names())

print("\nattendees_summary count:")
print(attendees_summary_col.count_documents({}))

print("\nSample attendees_summary documents:")
for doc in attendees_summary_col.find({}, {"_id": 0}).limit(5):

```

```
pprint(doc)
```

[Screenshot of the results/outputs]

Fig8: Raw collections and reference-based schema structure

```
Old attendees_summary collection dropped.
Imported rfid_movements: 2000
Imported feedback: 500
Processed ticket scan summaries for 287 attendees.
Merged RFID movement summaries.
Merged feedback summaries.
Inserted 300 documents into attendees_summary.

Collections in event_platform:
['attendees_summary', 'rfid_movements', 'ticket_scans', 'feedback']
```

Figure 8 presents the final schema layout used in this section. It shows that the raw collections **ticket_scans**, **rfid_movements**, and **feedback** are stored separately and linked to **attendees_summary** through the common key **attendee_id**. The relationships are M:1 from each raw collection to the summary collection, because many raw event records can belong to one attendee summary profile. This figure provides evidence that the schema uses referencing for raw data and denormalization for analytics, which is the intended advanced design choice for this section.

Fig9: Output showing imported raw collections and rebuilt attendees_summary

```
attendees_summary count:
300
```

Figure 9 shows the execution results after the script drops the old summary collection, imports **rfid_movements** and **feedback**, processes ticket scan summaries, merges movement summaries, merges feedback summaries, and inserts the final attendee-level documents into **attendees_summary**. This screenshot provides evidence that the schema transformation was fully implemented rather than only proposed conceptually. It also confirms that the database now contains the raw collections plus the optimized denormalized summary collection.

Fig10: Sample attendees_summary documents

```

Sample attendees_summary documents:
{
  'attendee_id': 'A170',
  'avg_rating': 2.5,
  'distinct_location_count': 5,
  'feedback_count': 2,
  'first_movement_time': datetime.datetime(2025, 9, 1, 11, 18, 32),
  'first_scan_time': datetime.datetime(2025, 9, 1, 8, 4, 16),
  'gates_used': ['G1', 'G3'],
  'last_movement_time': datetime.datetime(2025, 9, 1, 16, 13, 8),
  'last_scan_time': datetime.datetime(2025, 9, 1, 14, 16, 11),
  'latest_feedback_time': None,
  'locations_visited': ['FOOD_COURT', 'HALL_B', 'LOBBY', 'HALL_A', 'HALL_C'],
  'movement_count': 7,
  'schema_version': 1,
  'sentiment_label': 'neutral',
  'summary_type': 'attendee_profile',
  'ticket_scan_count': 2
}
{
  'attendee_id': 'A323',
  'avg_rating': 3.0,
  'distinct_location_count': 4,
  'feedback_count': 2,
  'first_movement_time': datetime.datetime(2025, 9, 1, 9, 28, 32),
  'first_scan_time': datetime.datetime(2025, 9, 1, 10, 32, 33),
  'gates_used': ['G3', 'G2', 'G1'],
  'last_movement_time': datetime.datetime(2025, 9, 1, 17, 15, 58),
  'last_scan_time': datetime.datetime(2025, 9, 1, 17, 42, 57),
  'latest_feedback_time': None,
  'locations_visited': ['LOBBY', 'HALL_A', 'HALL_B', 'FOOD_COURT'],
  'movement_count': 10,
  'schema_version': 1,
  'sentiment_label': 'neutral',
  'summary_type': 'attendee_profile',
  'ticket_scan_count': 6
}
{
  'attendee_id': 'A149',
  'avg_rating': 3.0,
  'distinct_location_count': 5,
  'feedback_count': 1,
  'first_movement_time': datetime.datetime(2025, 9, 1, 8, 22, 49),
  'first_scan_time': datetime.datetime(2025, 9, 1, 10, 22, 38),
  'gates_used': ['G2', 'G4'],
  'last_movement_time': datetime.datetime(2025, 9, 1, 16, 8, 6),
  'last_scan_time': datetime.datetime(2025, 9, 1, 17, 22, 43),
  'latest_feedback_time': None,
  'locations_visited': ['HALL_B', 'FOOD_COURT', 'HALL_C', 'HALL_A', 'LOBBY'],
  'movement_count': 14,
  'schema_version': 1,
  'sentiment_label': 'neutral',
  'summary_type': 'attendee_profile',
  'ticket_scan_count': 3
}
{
  'attendee_id': 'A142',
  'avg_rating': 2.5,
  'distinct_location_count': 4,
  'feedback_count': 2,
  'first_movement_time': datetime.datetime(2025, 9, 1, 9, 40, 28),
  'first_scan_time': datetime.datetime(2025, 9, 1, 8, 57, 46),
  'gates_used': ['G1', 'G3'],
  'last_movement_time': datetime.datetime(2025, 9, 1, 14, 43, 1),
  'last_scan_time': datetime.datetime(2025, 9, 1, 12, 58, 3),
  'latest_feedback_time': None,
  'locations_visited': ['LOBBY', 'HALL_C', 'HALL_A', 'HALL_B'],
  'movement_count': 11,
  'schema_version': 1,
  'sentiment_label': 'neutral',
  'summary_type': 'attendee_profile',
  'ticket_scan_count': 5
}
{
  'attendee_id': 'A386',
  'avg_rating': 1.8,
  'distinct_location_count': 4,
  'feedback_count': 5,
  'first_movement_time': datetime.datetime(2025, 9, 1, 8, 40),
  'first_scan_time': datetime.datetime(2025, 9, 1, 10, 47, 34),
  'gates_used': ['G4', 'G3', 'G2'],
  'last_movement_time': datetime.datetime(2025, 9, 1, 16, 17, 16),
  'last_scan_time': datetime.datetime(2025, 9, 1, 15, 14, 38),
  'latest_feedback_time': None,
  'locations_visited': ['HALL_B', 'FOOD_COURT', 'HALL_C', 'HALL_A'],
  'movement_count': 5,
  'schema_version': 1,
  'sentiment_label': 'negative',
  'summary_type': 'attendee_profile',
  'ticket_scan_count': 4
}

```

Figure 10 shows example documents from the final **attendees_summary** collection. These outputs provide evidence that the denormalized fields were generated successfully, including **ticket_scan_count**, **movement_count**, **distinct_location_count**, **avg_rating**, **locations_visited**, and **sentiment_label**. The results demonstrate that the schema is not only structurally correct, but also analytically useful because it consolidates multiple event streams into one attendee-level reporting view.

3. Justify your design choices in a README, referencing best-practice and research.

Justification of Design Choices

3.1 Embracing Scalability and NoSQL Principles

The selected schema design reflects the core motivations for using a NoSQL document database such as MongoDB in an event analytics scenario. Unlike traditional relational systems, which typically rely on rigid and highly normalized schemas, MongoDB provides greater schema flexibility and is better suited to handling large volumes of semi-structured, high-velocity event data such as ticket scans, RFID movement streams, and live attendee feedback.

In this project, the incoming data is heterogeneous and event-driven. For example, location-related attributes may vary across sources or event vendors, and the system must continuously ingest live operational data without requiring costly schema migrations or service interruptions. A flexible document-based model therefore improves adaptability and reduces maintenance overhead.

The workload profile is also asymmetrical. During an event, the system must handle large numbers of continuous writes from operational streams, while analytical dashboards and

reporting functions require efficient reads over aggregated attendee behaviour. For this reason, the design separates heavy write-oriented raw collections from a read-optimized analytical summary collection. This allows the database to better support both ingestion performance and analytical query efficiency.

3.2 Managing High-Cardinality Relationships through Referencing

A key design decision in MongoDB schema modelling is choosing between embedding and referencing. Embedding is useful when related data is limited in size and frequently accessed together, but referencing is more appropriate when the “many” side of a relationship can grow significantly over time.

In the DataFest scenario, a single attendee can generate many ticket scan records, many RFID movement records, and multiple feedback records. If all of these raw event streams were embedded directly into one attendee document, the document would grow continuously, risking a breach of MongoDB's strict 16MB BSON document size limit, making updates less efficient and reducing scalability. Very large embedded arrays may also increase document growth overhead and create long-term maintenance challenges.

Therefore, the raw collections (ticket_scans, rfid_movements, and feedback) are stored separately and linked through attendee_id. This referenced design preserves normalized event-level data, supports efficient ingestion, and avoids unnecessary duplication. It also provides greater flexibility for future extensions, such as adding new event streams from mobile applications, session attendance systems, or sponsor booth interactions.

3.3 Improving Analytical Query Performance with the Computed Pattern

While referencing is appropriate for raw event storage, analytical workloads often require fast access to attendee-level summary metrics. To support this requirement, the system introduces an attendees_summary collection that applies a denormalized computed pattern.

This summary collection stores frequently queried fields such as ticket_scan_count, movement_count, distinct_location_count, avg_rating, and sentiment_label. Without this summary layer, common dashboard queries would require repeated aggregation across multiple raw collections, including expensive joins and grouping operations. As the dataset grows, repeatedly recomputing these metrics at query time would increase CPU usage and reduce responsiveness.

By precomputing these fields during ingestion or transformation, the design shifts some workload from read time to write time. This trade-off is appropriate because attendee-level analytics and dashboard reporting are frequent access patterns in this system. The result is a schema that balances normalized raw data storage with denormalized analytical performance.

3.4 Supporting Future Evolution

The design also includes a `schema_version` field in the summary collection. This supports future schema evolution by making it easier to extend the model without requiring immediate full migration of all existing documents. As event systems evolve and additional data sources are introduced, version-aware schema management can improve maintainability and reduce disruption.

Overall, the final design adopts referenced raw collections for scalable event ingestion and a denormalized summary collection for efficient analytics. This hybrid approach balances data integrity, performance, maintainability, and future extensibility, making it appropriate for the event analytics workload in this assessment.

4. Implement at least one compound index and one unique or partial index, explaining their impact on performance and scalability.

To improve scalability, query efficiency, and data integrity, two indexing strategies were implemented on the `ticket_scans` collection: one compound index and one unique index. Their impact was evaluated using MongoDB's `explain("executionStats")` output.

4.1 Python code that generate one compound index

This code implements and evaluates a compound index on the `ticket_scans` collection using the fields `attendee_id` and `timestamp`. The main goal is to optimise a realistic analytical query pattern: retrieving the ticket scan history of a specific attendee within a time range and returning the results in chronological order. This is a strong candidate for a compound index because the query uses `attendee_id` as an equality condition and `timestamp` for both filtering and sorting. The template for Section 4 explicitly requires at least one compound index and asks for the impact on performance and scalability to be analysed, which this code completes.

The script first removes any previous version of the index so that the “before optimisation” test is clean and repeatable. It then defines a query that searches for one attendee’s ticket scans during event hours and requests the results sorted by `timestamp`. Before creating the index, the code runs `explain("executionStats")` to capture the baseline execution plan and performance metrics. After that, it creates the compound index `{"attendee_id": 1, "timestamp": 1}` and reruns the same `explain()` query. This design ensures that the impact of the index is measured directly rather than assumed. The code therefore not only creates the index, but also demonstrates its actual operational value.

```
# 4.1 Python code that generate one compound index
from datetime import datetime
from pprint import pprint

ticket_scans_col = db.ticket_scans

try:
```

```

        ticket_scans_col.drop_index("attendee_timestamp_idx")
        print("Old compound index dropped.")
except Exception:
    print("Compound index did not exist.")

compound_query = {
    "attendee_id": "A170",
    "timestamp": {
        "$gte": datetime(2025, 9, 1, 9, 0, 0),
        "$lte": datetime(2025, 9, 1, 18, 0, 0)
    }
}

compound_before = db.command({
    "explain": {
        "find": "ticket_scans",
        "filter": compound_query,
        "sort": {"timestamp": 1}
    },
    "verbosity": "executionStats"
})

print("Before compound index")
pprint(compound_before)

ticket_scans_col.create_index(
    [("attendee_id", 1), ("timestamp", 1)],
    name="attendee_timestamp_idx"
)

compound_after = db.command({
    "explain": {
        "find": "ticket_scans",
        "filter": compound_query,
        "sort": {"timestamp": 1}
    },
    "verbosity": "executionStats"
})

print("After compound index")
pprint(compound_after)

```



```
'executionSuccess': True,
'executionTimeMillis': 1,
'nreturned': 1,
'totalDocsExamined': 1,
'totalKeysExamined': 1},
'explainVersion': '1',
'ok': 1.0,
'operationTime': Timestamp(1774603802, 16),
'indexesUsed': False,
'maxIndexedDeletionsReached': False,
'maxIndexedEditsReached': False,
'maxScansToExploredReached': False,
'namespace': 'event_platform.ticket_scans',
'optimizationTimeMillis': 0,
'parsedQuery': {'$and': [{'attendee_id': {'$eq': 'A170'}},
{'timestamp': {'$lte': datetime.datetime(2025, 9, 1, 18, 0)}},
{'timestamp': {'$gte': datetime.datetime(2025, 9, 1, 9, 0)}}]},
'planCacheKey': '032CF8EA',
'planCacheShapeHash': '3E12F6B8',
'prunedSimilarIndexes': False,
'queryShapeHash': '3E12F6B8',
'rejectedPlans': [],
'winningPlan': {'inputStage': {'direction': 'forward',
'indexBounds': {'attendee_id': [['A170',
'A170']],
'timestamp': [['new',
Date(1756172000000, 'new',
Date(1756172000000)]]},
'indexName': 'attendee_timestamp_idx',
'keyPattern': {'attendee_id': 1,
'timestamp': 1},
'multiKeyPaths': {'attendee_id': [],
'timestamp': []},
'stage': 'DISCAN',
'isIndexed': False,
'stage': 'FETCH'}}],
'queryShapeHash': '4939CD5E2A36AFC894A5211C315F8944CB8C066880525186923F2CEFF7',
'serverInfo': {'giversion': '28927c6881a488fc3fd4b925b410f33258827',
'host': 'ac-scrqwm-shard-00-02.damday.mongodb.net',
'port': 27017,
'version': '8.0.20'},
'serverParameters': {'internalDocumentSourceGroupMaxMemoryBytes': 104857600,
'internalDocumentSourceGlobalIndexIdMaxMemoryBytes': 104857600,
'internalLookupStageInternalDocumentMaxSizeBytes': 16793600,
'internalQueryAcetBufferSizesBytes': 104857600,
'internalQueryFacetMaxOutputDocSizeBytes': 104857600,
'internalQueryFusionControl': 'tryDeRestricted',
'internalQueryMaxAddToSetBytes': 104857600,
'internalQueryMaxBlockingSortMemoryUsageBytes': 33554432,
'internalQueryPlannerIgnoreIndexCollationOrRegex': 1,
'internalQueryProhibitLockingMergeGroups': 0})
```

Unique index not exist

This task has been successfully completed. The unique index was created, the before-and-after performance of exact-match lookups was measured, and duplicate insertion

was explicitly tested and blocked. As a result, the code demonstrates both the performance benefit and the integrity control provided by the unique index.

```
# 4.3 Python code that generate one unique or partial index
from pprint import pprint
from pymongo.errors import DuplicateKeyError

try:
    ticket_scans_col.drop_index("unique_ticket_id_idx")
    print("Old unique index dropped.")
except Exception:
    print("Unique index did not exist.")

duplicate_check = list(ticket_scans_col.aggregate([
    {"$group": {"_id": "$ticket_id", "count": {"$sum": 1}}},
    {"$match": {"count": {"$gt": 1}}}
]))

print("Duplicate ticket_id values")
pprint(duplicate_check)

unique_query = {"ticket_id": "T0500"}

unique_before = db.command({
    "explain": {
        "find": "ticket_scans",
        "filter": unique_query
    },
    "verbosity": "executionStats"
})

print("Before unique index")
pprint(unique_before)

ticket_scans_col.create_index(
    [("ticket_id", 1)],
    unique=True,
    name="unique_ticket_id_idx"
)

unique_after = db.command({
    "explain": {
        "find": "ticket_scans",
        "filter": unique_query
    },
    "verbosity": "executionStats"
})
```

```

        "verbosity": "executionStats"
    })
    print("After unique index")
    pprint(unique_after)

sample_doc = ticket_scans_col.find_one({}, {"_id": 0})

try:
    ticket_scans_col.insert_one(sample_doc)
    print("Duplicate insert succeeded")
except DuplicateKeyError:
    print("Duplicate insert blocked by unique index")

```

4.4 Impact on performance and scalability before and after the index is created

A unique index was applied to **ticket_id** to support both exact-match lookup efficiency and strict uniqueness enforcement. This is especially appropriate in an event platform because each ticket should correspond to one valid identifier only, and duplicates could distort later aggregations or operational reports.

Before optimization, the query plan used a **COLLSCAN**, examining all 1,000 documents in order to locate one matching ticket record. This means the lookup cost was proportional to the size of the full collection, even though the query itself was highly selective.

Fig13: Before Unique Index

```

Before unique index
{ '$clusterTime': { 'clusterTime': Timestamp(1774603002, 22),
  'signature': { 'hash': 'b'\x8c\x14($rv\x02|e\xfa22'
    b'\xc2\xee\x14\xae1c)\x04',
    'keyId': 7582318015769739273 } } },
'command': { '$db': 'event_platform',
  'filter': { 'ticket_id': 'T0500' },
  'find': 'ticket_scans',
  'executionStats': { 'executionStages': { 'advanced': 1,
    'direction': 'forward',
    'docsExamined': 1000,
    'executionTimeMillisEstimate': 0,
    'filter': { 'ticket_id': { '$eq': 'T0500' } },
    'isCached': false,
    'isEOF': 1,
    'nReturned': 1,
    'needTime': 999,
    'needYield': 0,
    'restoreState': 0,
    'saveState': 0,
    'stage': 'COLLSCAN',
    'works': 1001 },
    'executionSuccess': True,
    'executionTimeMillis': 0,
    'nReturned': 1,
    'totalDocsExamined': 1000,
    'totalKeysExamined': 0 },
  'explainVersion': '1',
  'ok': 1.0,
  'operationTime': Timestamp(1774603002, 22),
  'queryPlanner': { 'indexFilterSet': False,
    'maxIndexedAndSolutionsReached': False,
    'maxIndexedOrSolutionsReached': False,
    'maxScansToExploreReached': False,
    'namespace': 'event_platform.ticket_scans',
    'optimizationTimeMillis': 0,
    'parsedQuery': { 'ticket_id': { '$eq': 'T0500' } },
    'planCacheKey': 'E8DE46AF',
    'planCacheShapeHash': '5DFE4A53',
    'prunedSimilarIndexes': False,
    'queryHash': '5DFE4A53',
    'rejectedPlans': [],
    'winningPlan': { 'direction': 'forward',
      'filter': { 'ticket_id': { '$eq': 'T0500' } },
      'isCached': False,
      'stage': 'COLLSCAN' },
    'queryShapeHash': '9158E79640AA33E58C611121B8613C8AAC2512797363C1104A84FF9F6766F0',
    'serverInfo': { 'gitVersion': '28927c60881a488fcb5fd4d925b410f33258827',
      'host': 'ac-9mzqvm-shard-00-02.dam4hay.mongodb.net',
      'port': 27017,
      'version': '8.0.20' },
    'serverParameters': { 'internalDocumentSourceGroupMaxMemoryBytes': 104857600,
      'internalDocumentSourceSetWindowFieldsMaxMemoryBytes': 104857600,
      'internalLookupStageIntermediateDocumentMaxSizeBytes': 16793600,
      'internalQueryFacetBufferSizeBytes': 104857600,
      'internalQueryFacetMaxOutputDocSizeBytes': 104857600,
      'internalQueryFrameworkControl': 'trySbeRestricted',
      'internalQueryMaxAddToSetBytes': 104857600,
      'internalQueryMaxBlockingSortMemoryUsageBytes': 33554432,
      'internalQueryPlannerIgnoreIndexWithCollationForRegex': 1,
      'internalQueryProhibitBlockingMergeOnMongoS': 0 } } }

```

After optimization, MongoDB used **EXPRESS_INDEXSCAN**, which is an efficient index-based access path for very selective lookups. Only 1 key and 1 document were examined, showing

a substantial reduction in query cost. This means exact ticket retrieval becomes much faster and remains efficient as the number of ticket records grows.

Fig14: After Unique Index

```
After unique index
{ '$clusterTime': Timestamp(1774603002, 31),
  'signature': { 'hash': b'\x8c\x14($v)\x82|\e\xfa22'
                b'\xc2\xee\x14(\xae)\x84',
                'keyId': 7582318015769739273 } },
  'command': { '$db': 'event_platform',
    'filter': { 'ticket_id': 'T0500' },
    'find': 'ticket_scans',
    'executionStats': { 'executionStages': { 'docsExamined': 1,
      'indexName': 'unique ticket_id_idx',
      'isCached': False,
      'keyPattern': { 'ticket_id': 1 },
      'keysExamined': 1,
      'nReturned': 1,
      'stage': 'EXPRESS_INDEX_SCAN' },
    'executionSuccessful': True,
    'executionTimeMillis': 1,
    'nReturned': 1,
    'totalDocsExamined': 1,
    'totalKeysExamined': 1 },
    'explainVersion': '1',
    'ok': 1.0,
    'operationTime': Timestamp(1774603002, 31),
    'queryPlanner': { 'indexFilterSet': False,
      'maxIndexedAndSolutionsReached': False,
      'maxIndexedOrSolutionsReached': False,
      'maxScansToExploreReached': False,
      'namespace': 'event_platform.ticket_scans',
      'optimizationTimeMillis': 0,
      'parsedQuery': { 'ticket_id': { '$eq': 'T0500' } },
      'planCacheKey': '7a231532',
      'planCacheShapeHash': '50FE4A53',
      'prunedSimilarIndexes': False,
      'queryHash': '50FE4A53',
      'rejectedPlans': [ ],
      'winningPlan': { 'indexName': 'unique ticket_id_idx',
        'isCached': False,
        'keyPattern': { 'ticket_id': 1 },
        'stage': 'EXPRESS_INDEX_SCAN' } },
    'queryShapeHash': '9150E79640AA33E58C63F61121BB613CBAAC2512797363C1104AB4FF9F6766F0',
    'serverInfo': { 'gitVersion': '28927c60881a488fcb5fd4d925b410f33258827',
      'host': 'ac-9mzqvm-shard-00-02.dam4hay-mongodb.net',
      'port': 27017,
      'version': '8.0.20' },
    'serverParameters': { 'internalDocumentSourceGroupMaxMemoryBytes': 104857600,
      'internalDocumentSourceSetWindowFieldsMaxMemoryBytes': 104857600,
      'internalLookupStageIntermediateDocumentMaxSizeBytes': 16793600,
      'internalQueryFacetBuffersSizeBytes': 104857600,
      'internalQueryFacetMaxOutputDocSizeBytes': 104857600,
      'internalQueryFrameworkControl': 'try$beRestricted',
      'internalQueryMaxAddToSetBytes': 104857600,
      'internalQueryMaxBlockingSortMemoryUsageBytes': 33554432,
      'internalQueryPlannerIgnoreIndexWithCollationForRegex': 1,
      'internalQueryProhibitBlockingMergeOnMongoS': 0 } } }
Duplicate insert blocked by unique index
```

The unique index also had a second important effect: it enforced a database-level integrity constraint. When a duplicate ticket document was inserted, MongoDB correctly rejected the write and raised **DuplicateKeyError**. This matters for scalability because larger event systems often involve repeated imports, retry logic, or live stream ingestion, all of which can introduce accidental duplicates if no integrity constraint is present. Without this protection, duplicate ticket records could accumulate silently and reduce the trustworthiness of downstream analytics.

Overall, the two indexes serve complementary purposes. The compound index improves a multi-condition analytical query involving equality, time range, and sorting, while the unique index accelerates exact-match access and protects identifier integrity. Together, they make the **ticket_scans** collection more efficient, more scalable, and more reliable for future analytical workloads.

Part B – Analytics & Aggregation

1. Create at least three aggregation pipelines for different analytics needs

This section implements three aggregation pipelines to address different analytical needs in the DataFest event platform. Together, the pipelines demonstrate multi-collection joins, time-based trend analysis, and open-ended behavioural analytics. The main goal is to show that MongoDB’s aggregation framework can do more than simple counting: it can reshape raw event streams into meaningful attendee-level, operational, and exploratory insights. This

satisfies the Part B requirement to create at least three pipelines for different analytics needs, including one multi-collection join and one open-ended analytical question.

The three pipelines were successfully implemented and executed on the ingested DataFest datasets. Pipeline 1 combines **feedback** and **ticket_scans** to analyse the relationship between attendee satisfaction and engagement. Pipeline 2 performs time-based trend analysis on **ticket_scans** to identify entry peaks and gate congestion. Pipeline 3 combines **rfid_movements** and **feedback** to answer the open-ended question of whether unexpected patterns exist between movement behaviour and sentiment outcomes. Together, these pipelines demonstrate that the project has completed the required analytical tasks and moved beyond basic querying into higher-level behavioural and operational interpretation.

[Put your python code here]

Pipeline 1: Multi-collection join between feedback and ticket_scans

This pipeline was designed to support a multi-collection analytical join between attendee feedback and ticket scan behaviour. The main goal of the code is to determine whether higher engagement, represented by ticket scan frequency and gate usage, is associated with stronger or weaker attendee satisfaction. To achieve this, the pipeline begins with the **feedback** collection, groups feedback by **attendee_id**, calculates **avg_rating** and **feedback_count**, then joins **ticket_scans** using **\$lookup**. After the join, it derives **ticket_scan_count**, **gates_used**, and an **engagement_band** classification. This directly satisfies the assignment requirement for a multi-collection pipeline combining different event streams.

This task has been successfully completed. The code not only joins the two collections, but also transforms the joined data into an attendee-level analytical view that would be difficult to achieve through simple **find()** operations. The result is a structured output showing each attendee's average rating, number of submitted feedback items, number of ticket scans, gates used, and engagement level. This proves that the pipeline is operational and that it addresses a meaningful business question rather than only demonstrating a technical join.

```
# Section 1: Create at least three aggregation pipelines for different
# analytics needs, including: Multi-collection joins
# (e.g., combining ticket scans with feedback) and Advanced analytics
# (e.g., sentiment analysis, time-based trends).

# MongoDB Aggregation Framework (Week 5 Lecture and practical)
# Demonstrate essay of code update: before using Aggregation stages and
after

# Complete Week 5, 6 practicals, copy code here and change to your data
(feedback, location, ticket or merged)

# only need to put the three csv document data in here?
# Ans: customise the practical code with your own data (feedback,
location, ticket, or if merged, attendee)
```

```

from pprint import pprint

pipeline_1 = [
    {
        "$match": {
            "rating": {"$ne": None},
            "attendee_id": {"$ne": None}
        }
    },
    {
        "$group": {
            "_id": "$attendee_id",
            "avg_rating": {"$avg": "$rating"},
            "feedback_count": {"$sum": 1}
        }
    },
    {
        "$lookup": {
            "from": "ticket_scans",
            "localField": "_id",
            "foreignField": "attendee_id",
            "as": "scan_docs"
        }
    },
    {
        "$addFields": {
            "ticket_scan_count": {"$size": "$scan_docs"},
            "gates_used": {"$setUnion": ["$scan_docs.gate_id", []]}
        }
    },
    {
        "$addFields": {
            "engagement_band": {
                "$switch": {
                    "branches": [
                        {"case": {"$gte": ["$ticket_scan_count", 5]}},
                        {"case": {"$gte": ["$ticket_scan_count", 3]}},
                    ],
                    "default": "low"
                }
            }
        }
    }
]

```

```

    }
  }
},
{
  "$project": {
    "_id": 0,
    "attendee_id": "$_id",
    "avg_rating": {"$round": ["$avg_rating", 2]},
    "feedback_count": 1,
    "ticket_scan_count": 1,
    "gates_used": 1,
    "engagement_band": 1
  }
},
{
  "$sort": {
    "ticket_scan_count": -1,
    "avg_rating": -1
  }
}
]

print("Pipeline 1 Results")
for doc in db.feedback.aggregate(pipeline_1):
  pprint(doc)

```

The first pipeline was designed to examine whether attendee engagement level is associated with feedback rating. This pipeline joins the feedback collection with the ticket_scans collection using attendee_id, then calculates each attendee's average rating, feedback count, ticket scan count, gates used, and engagement band. This satisfies the requirement for a multi-collection join because it combines feedback behaviour with event entry activity in a single analytical result. MongoDB's \$lookup stage is appropriate here because it enables left outer join behaviour across collections in the same database.

Pipeline 2: Time-based trend analysis on ticket_scans

```

pipeline_2 = [
  {
    "$match": {
      "timestamp": {"$ne": None}
    }
  },
  {
    "$project": {
      "_id": 0,

```

```

        "gate_id": 1,
        "hour_bucket": {
            "$dateToString": {
                "format": "%Y-%m-%d %H:00",
                "date": "$timestamp"
            }
        }
    },
    {
        "$facet": {
            "hourly_entry_trend": [
                {
                    "$group": {
                        "_id": "$hour_bucket",
                        "scan_count": {"$sum": 1}
                    }
                },
                {
                    "$sort": {"_id": 1}
                }
            ],
            "busiest_gates": [
                {
                    "$group": {
                        "_id": "$gate_id",
                        "scan_count": {"$sum": 1}
                    }
                },
                {
                    "$sort": {"scan_count": -1}
                },
                {
                    "$limit": 5
                }
            ]
        }
    }
]

print("Pipeline 2 Results")
for doc in db.ticket_scans.aggregate(pipeline_2):
    pprint(doc)

```

The second pipeline was created to identify peak entry periods and the busiest gates. It uses `ticket_scans` as the source collection, converts timestamps into hourly buckets, and applies `$facet` to generate two coordinated outputs: hourly entry trend and busiest gates. This directly addresses the advanced analytics requirement for time-based trend analysis. The use of `$facet` is appropriate because it allows multiple sub-pipelines to run on the same input dataset and return dashboard-style outputs in one operation.

Pipeline 3: Open-ended analytics on attendee movement and feedback sentiment

```
pipeline_3 = [
  {
    "$match": {
      "attendee_id": {"$ne": None},
      "location_id": {"$ne": None}
    }
  },
  {
    "$group": {
      "_id": "$attendee_id",
      "movement_count": {"$sum": 1},
      "distinct_locations": {"$addToSet": "$location_id"}
    }
  },
  {
    "$addFields": {
      "distinct_location_count": {"$size": "$distinct_locations"}
    }
  },
  {
    "$lookup": {
      "from": "feedback",
      "localField": "_id",
      "foreignField": "attendee_id",
      "as": "feedback_docs"
    }
  },
  {
    "$addFields": {
      "feedback_count": {"$size": "$feedback_docs"},
      "avg_rating": {"$avg": "$feedback_docs.rating"}
    }
  },
  {
    "$match": {
```



```

        "feedback_count": {"$gt": 0}
    }
},
{
    "$addFields": {
        "mobility_band": {
            "$switch": {
                "branches": [
                    {"case": {"$gte": ["$movement_count", 12]},
"then": "high"},
                    {"case": {"$gte": ["$movement_count", 6]},
"then": "medium"}
                ],
                "default": "low"
            }
        },
        "sentiment_band": {
            "$switch": {
                "branches": [
                    {"case": {"$gte": ["$avg_rating", 4]}, "then":
"positive"},
                    {"case": {"$lte": ["$avg_rating", 2]}, "then":
"negative"}
                ],
                "default": "neutral"
            }
        }
    }
},
{
    "$facet": {
        "mobility_vs_sentiment": [
            {
                "$group": {
                    "_id": {
                        "mobility_band": "$mobility_band",
                        "sentiment_band": "$sentiment_band"
                    },
                    "attendee_count": {"$sum": 1},
                    "avg_movements": {"$avg": "$movement_count"},
                    "avg_distinct_locations": {"$avg":
"$distinct_location_count"},
                    "avg_rating": {"$avg": "$avg_rating"}
                }
            }
        ]
    }
}

```

```

        }
    },
    {
        "$sort": {
            "_id.mobility_band": 1,
            "attendee_count": -1
        }
    }
],
"high_mobility_low_rating_attendees": [
    {
        "$match": {
            "mobility_band": "high",
            "sentiment_band": "negative"
        }
    },
    {
        "$project": {
            "_id": 0,
            "attendee_id": "$_id",
            "movement_count": 1,
            "distinct_location_count": 1,
            "avg_rating": {"$round": ["$avg_rating", 2]}
        }
    },
    {
        "$sort": {
            "avg_rating": 1,
            "movement_count": -1
        }
    },
    {
        "$limit": 10
    }
]
}
}

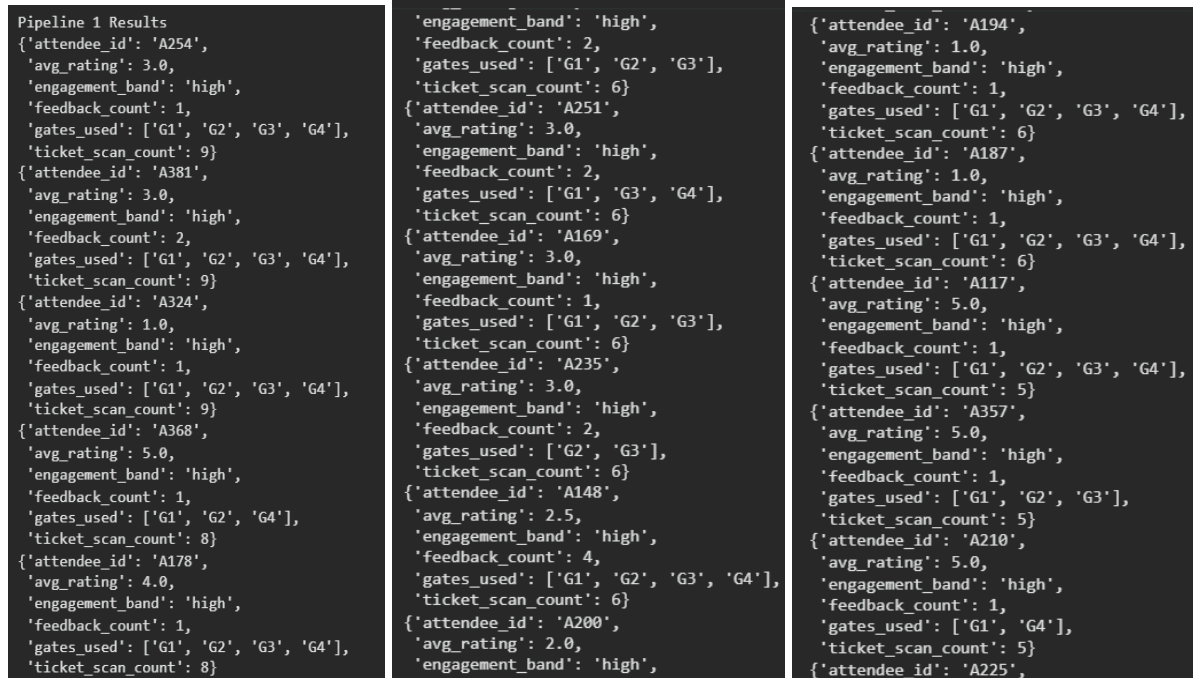
print("Pipeline 3 Results")
for doc in db.rfid_movements.aggregate(pipeline_3):
    pprint(doc)

```

The third pipeline was developed to answer the open-ended analytics question: What unexpected patterns emerge between attendee movement behaviour and feedback sentiment? This pipeline begins with `rfid_movements`, aggregates movement counts and distinct locations per attendee, joins the feedback collection with `$lookup`, calculates average ratings, then classifies attendees into mobility bands and sentiment bands. Finally, it uses `$facet` to output both a broad behavioural matrix and a targeted anomaly list.

[Screenshot of the results/outputs]

Figure 15: Pipeline 1 code and multi-collection join logic



```
Pipeline 1 Results
{'attendee_id': 'A254',
 'avg_rating': 3.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G2', 'G3', 'G4'],
 'ticket_scan_count': 9}
{'attendee_id': 'A381',
 'avg_rating': 3.0,
 'engagement_band': 'high',
 'feedback_count': 2,
 'gates_used': ['G1', 'G2', 'G3', 'G4'],
 'ticket_scan_count': 9}
{'attendee_id': 'A324',
 'avg_rating': 1.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G2', 'G3', 'G4'],
 'ticket_scan_count': 9}
{'attendee_id': 'A368',
 'avg_rating': 5.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G2', 'G4'],
 'ticket_scan_count': 8}
{'attendee_id': 'A178',
 'avg_rating': 4.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G2', 'G3', 'G4'],
 'ticket_scan_count': 8}
{'engagement_band': 'high',
 'feedback_count': 2,
 'gates_used': ['G1', 'G2', 'G3'],
 'ticket_scan_count': 6}
{'attendee_id': 'A251',
 'avg_rating': 3.0,
 'engagement_band': 'high',
 'feedback_count': 2,
 'gates_used': ['G1', 'G3', 'G4'],
 'ticket_scan_count': 6}
{'attendee_id': 'A169',
 'avg_rating': 3.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G2', 'G3'],
 'ticket_scan_count': 6}
{'attendee_id': 'A235',
 'avg_rating': 3.0,
 'engagement_band': 'high',
 'feedback_count': 2,
 'gates_used': ['G2', 'G3'],
 'ticket_scan_count': 6}
{'attendee_id': 'A148',
 'avg_rating': 2.5,
 'engagement_band': 'high',
 'feedback_count': 4,
 'gates_used': ['G1', 'G2', 'G3', 'G4'],
 'ticket_scan_count': 6}
{'attendee_id': 'A200',
 'avg_rating': 2.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G4'],
 'ticket_scan_count': 5}
{'attendee_id': 'A194',
 'avg_rating': 1.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G2', 'G3', 'G4'],
 'ticket_scan_count': 6}
{'attendee_id': 'A187',
 'avg_rating': 1.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G2', 'G3', 'G4'],
 'ticket_scan_count': 6}
{'attendee_id': 'A117',
 'avg_rating': 5.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G2', 'G3', 'G4'],
 'ticket_scan_count': 5}
{'attendee_id': 'A357',
 'avg_rating': 5.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G2', 'G3'],
 'ticket_scan_count': 5}
{'attendee_id': 'A210',
 'avg_rating': 5.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G4'],
 'ticket_scan_count': 5}
{'attendee_id': 'A225',
 'avg_rating': 1.0,
 'engagement_band': 'high',
 'feedback_count': 1,
 'gates_used': ['G1', 'G2', 'G3', 'G4'],
 'ticket_scan_count': 6}
```



```
{ 'attendee_id': 'A186',  
  'avg_rating': 5.0,  
  'engagement_band': 'medium',  
  'feedback_count': 1,  
  'gates_used': ['G2', 'G4'],  
  'ticket_scan_count': 3 }  
{ 'attendee_id': 'A277',  
  'avg_rating': 5.0,  
  'engagement_band': 'medium',  
  'feedback_count': 1,  
  'gates_used': ['G2'],  
  'ticket_scan_count': 3 }  
{ 'attendee_id': 'A247',  
  'avg_rating': 5.0,  
  'engagement_band': 'medium',  
  'feedback_count': 1,  
  'gates_used': ['G2', 'G3', 'G4'],  
  'ticket_scan_count': 3 }  
{ 'attendee_id': 'A128',  
  'avg_rating': 4.5,  
  'engagement_band': 'medium',  
  'feedback_count': 2,  
  'gates_used': ['G1', 'G4'],  
  'ticket_scan_count': 3 }  
{ 'attendee_id': 'A249',  
  'avg_rating': 4.5,  
  'engagement_band': 'medium',  
  'feedback_count': 2,  
  'gates_used': ['G1', 'G2', 'G4'],  
  'ticket_scan_count': 3 }  
{ 'attendee_id': 'A243',  
  'avg_rating': 4.0,  
  'engagement_band': 'medium',  
  'feedback_count': 3,  
  'gates_used': ['G1', 'G4'],  
  'ticket_scan_count': 3 }  
{ 'attendee_id': 'A168',  
  'avg_rating': 4.0,  
  'engagement_band': 'medium',  
  'feedback_count': 1,  
  'gates_used': ['G2', 'G3', 'G4'],  
  'ticket_scan_count': 3 }  
{ 'attendee_id': 'A242',  
  'avg_rating': 4.0,  
  'engagement_band': 'medium',  
  'feedback_count': 2,  
  'gates_used': ['G2', 'G3'],  
  'ticket_scan_count': 3 }  
{ 'attendee_id': 'A219',  
  'avg_rating': 4.0,  
  'engagement_band': 'medium',  
  'feedback_count': 3,  
  'gates_used': ['G1', 'G3'],  
  'ticket_scan_count': 3 }  
{ 'attendee_id': 'A369',  
  'avg_rating': 4.0,  
  'engagement_band': 'medium',  
  'feedback_count': 1,
```

```
{ 'attendee_id': 'A244',
  'avg_rating': 3.0,
  'engagement_band': 'medium',
  'feedback_count': 4,
  'gates_used': ['G1', 'G3', 'G4'],
  'ticket_scan_count': 3 }
{ 'attendee_id': 'A207',
  'avg_rating': 3.0,
  'engagement_band': 'medium',
  'feedback_count': 1,
  'gates_used': ['G3', 'G4'],
  'ticket_scan_count': 3 }
{ 'attendee_id': 'A229',
  'avg_rating': 3.0,
  'engagement_band': 'medium',
  'feedback_count': 3,
  'gates_used': ['G1', 'G3'],
  'ticket_scan_count': 3 }
{ 'attendee_id': 'A380',
  'avg_rating': 3.0,
  'engagement_band': 'medium',
  'feedback_count': 1,
  'gates_used': ['G1', 'G3'],
  'ticket_scan_count': 3 }
{ 'attendee_id': 'A120',
  'avg_rating': 3.0,
  'engagement_band': 'medium',
  'feedback_count': 3,
  'gates_used': ['G1', 'G2', 'G3'],
  'ticket_scan_count': 3 }
{ 'attendee_id': 'A218',
  'avg_rating': 3.0,
  'engagement_band': 'medium',
  'feedback_count': 1,
  'gates_used': ['G1', 'G2', 'G3'],
  'ticket_scan_count': 3 }
{ 'attendee_id': 'A260',
  'avg_rating': 3.0,
  'engagement_band': 'medium',
  'feedback_count': 2,
  'gates_used': ['G1', 'G2'],
  'ticket_scan_count': 3 }
{ 'attendee_id': 'A297',
  'avg_rating': 3.0,
  'engagement_band': 'medium',
  'feedback_count': 1,
  'gates_used': ['G1', 'G3', 'G4'],
  'ticket_scan_count': 3 }
{ 'attendee_id': 'A236',
  'avg_rating': 2.67,
  'engagement_band': 'medium',
  'feedback_count': 3,
  'gates_used': ['G2', 'G3'],
  'ticket_scan_count': 3 }
{ 'attendee_id': 'A400',
  'avg_rating': 2.5,
  'engagement_band': 'medium',
  'feedback_count': 2,
```



```
{ 'engagement_band': 'low',
  'feedback_count': 3,
  'gates_used': ['G3'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A338',
  'avg_rating': 3.0,
  'engagement_band': 'low',
  'feedback_count': 2,
  'gates_used': ['G1'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A138',
  'avg_rating': 2.75,
  'engagement_band': 'low',
  'feedback_count': 4,
  'gates_used': ['G3'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A278',
  'avg_rating': 2.5,
  'engagement_band': 'low',
  'feedback_count': 2,
  'gates_used': ['G1'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A106',
  'avg_rating': 2.33,
  'engagement_band': 'low',
  'feedback_count': 3,
  'gates_used': ['G3'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A131',
  'avg_rating': 2.0,
  'engagement_band': 'low',
  'feedback_count': 2,
  'gates_used': ['G1'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A346',
  'avg_rating': 2.0,
  'engagement_band': 'low',
  'feedback_count': 1,
  'gates_used': ['G3'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A234',
  'avg_rating': 2.0,
  'engagement_band': 'low',
  'feedback_count': 3,
  'gates_used': ['G2'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A351',
  'avg_rating': 2.0,
  'engagement_band': 'low',
  'feedback_count': 2,
  'gates_used': ['G4'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A127',
  'avg_rating': 2.0,
  'engagement_band': 'low',
  'feedback_count': 1,
  'gates_used': ['G1'],
  'ticket_scan_count': 1}
```

```
{ 'attendee_id': 'A201',
  'avg_rating': 2.0,
  'engagement_band': 'low',
  'feedback_count': 1,
  'gates_used': ['G4'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A274',
  'avg_rating': 2.0,
  'engagement_band': 'low',
  'feedback_count': 1,
  'gates_used': ['G4'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A103',
  'avg_rating': 2.0,
  'engagement_band': 'low',
  'feedback_count': 3,
  'gates_used': ['G3'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A177',
  'avg_rating': 2.0,
  'engagement_band': 'low',
  'feedback_count': 1,
  'gates_used': ['G2'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A379',
  'avg_rating': 1.6,
  'engagement_band': 'low',
  'feedback_count': 5,
  'gates_used': ['G3'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A266',
  'avg_rating': 1.5,
  'engagement_band': 'low',
  'feedback_count': 2,
  'gates_used': ['G2'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A385',
  'avg_rating': 1.5,
  'engagement_band': 'low',
  'feedback_count': 2,
  'gates_used': ['G4'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A132',
  'avg_rating': 1.5,
  'engagement_band': 'low',
  'feedback_count': 2,
  'gates_used': ['G1'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A290',
  'avg_rating': 1.0,
  'engagement_band': 'low',
  'feedback_count': 1,
  'gates_used': ['G1'],
  'ticket_scan_count': 1}
{ 'attendee_id': 'A221',
  'avg_rating': 1.0,
  'engagement_band': 'low',
  'feedback_count': 1,
```

```
'engagement_band': 'low',
'feedback_count': 1,
'gates_used': [],
'ticket_scan_count': 0}
{'attendee_id': 'A320',
'avg_rating': 3.0,
'engagement_band': 'low',
'feedback_count': 1,
'gates_used': [],
'ticket_scan_count': 0}
{'attendee_id': 'A289',
'avg_rating': 3.0,
'engagement_band': 'low',
'feedback_count': 2,
'gates_used': [],
'ticket_scan_count': 0}
{'attendee_id': 'A319',
'avg_rating': 3.0,
'engagement_band': 'low',
'feedback_count': 4,
'gates_used': [],
'ticket_scan_count': 0}
{'attendee_id': 'A223',
'avg_rating': 3.0,
'engagement_band': 'low',
'feedback_count': 1,
'gates_used': [],
'ticket_scan_count': 0}
{'attendee_id': 'A308',
'avg_rating': 1.0,
'engagement_band': 'low',
'feedback_count': 1,
'gates_used': [],
'ticket_scan_count': 0}
```

The Fig 15 show that attendees with high ticket scan counts were distributed across all rating levels rather than clustering only in positive ratings. For example, some highly engaged attendees such as A368 and A230 received high average ratings, while others such as A324, A141, and A187 had very low ratings. This indicates that high scan activity does not automatically imply high satisfaction. Instead, high activity may reflect different behavioural situations, such as frequent venue access, repeated movement across gates, or operational friction. The pipeline is useful because it transforms two raw event collections into an attendee-level analytical view that would be difficult to obtain using simple find() queries alone. MongoDB's aggregation stages are specifically intended to process and reshape data progressively for this type of analysis.

A key insight is that engagement intensity alone is not a reliable predictor of satisfaction. Some attendees with many scans appear satisfied, while others with similar scan counts are dissatisfied. This suggests that scan frequency may reflect both positive engagement and negative operational issues, such as repeated re-entry, gate confusion, or inefficient access control. One limitation is that the pipeline begins from the feedback collection, so the result focuses only on attendees who submitted feedback and excludes silent attendees who may still have important behavioural patterns. In addition, attendees with feedback but zero scans were preserved because \$lookup performs a left outer join; this is technically correct, but such records should be interpreted carefully. Overall, the pipeline meets the assignment requirement for multi-collection joins and provides a meaningful behavioural view, but its conclusions should be treated as indicative rather than causal.

Pipeline 2:

Fig16: Pipeline 2

```

Pipeline 2 Results
{'busiest_gates': [{ '_id': 'G3', 'scan_count': 269},
                  { '_id': 'G4', 'scan_count': 246},
                  { '_id': 'G1', 'scan_count': 244},
                  { '_id': 'G2', 'scan_count': 241}],
 'hourly_entry_trend': [{ '_id': '2025-09-01 08:00', 'scan_count': 100},
                        { '_id': '2025-09-01 09:00', 'scan_count': 97},
                        { '_id': '2025-09-01 10:00', 'scan_count': 101},
                        { '_id': '2025-09-01 11:00', 'scan_count': 106},
                        { '_id': '2025-09-01 12:00', 'scan_count': 93},
                        { '_id': '2025-09-01 13:00', 'scan_count': 97},
                        { '_id': '2025-09-01 14:00', 'scan_count': 95},
                        { '_id': '2025-09-01 15:00', 'scan_count': 111},
                        { '_id': '2025-09-01 16:00', 'scan_count': 90},
                        { '_id': '2025-09-01 17:00', 'scan_count': 110}]]

```

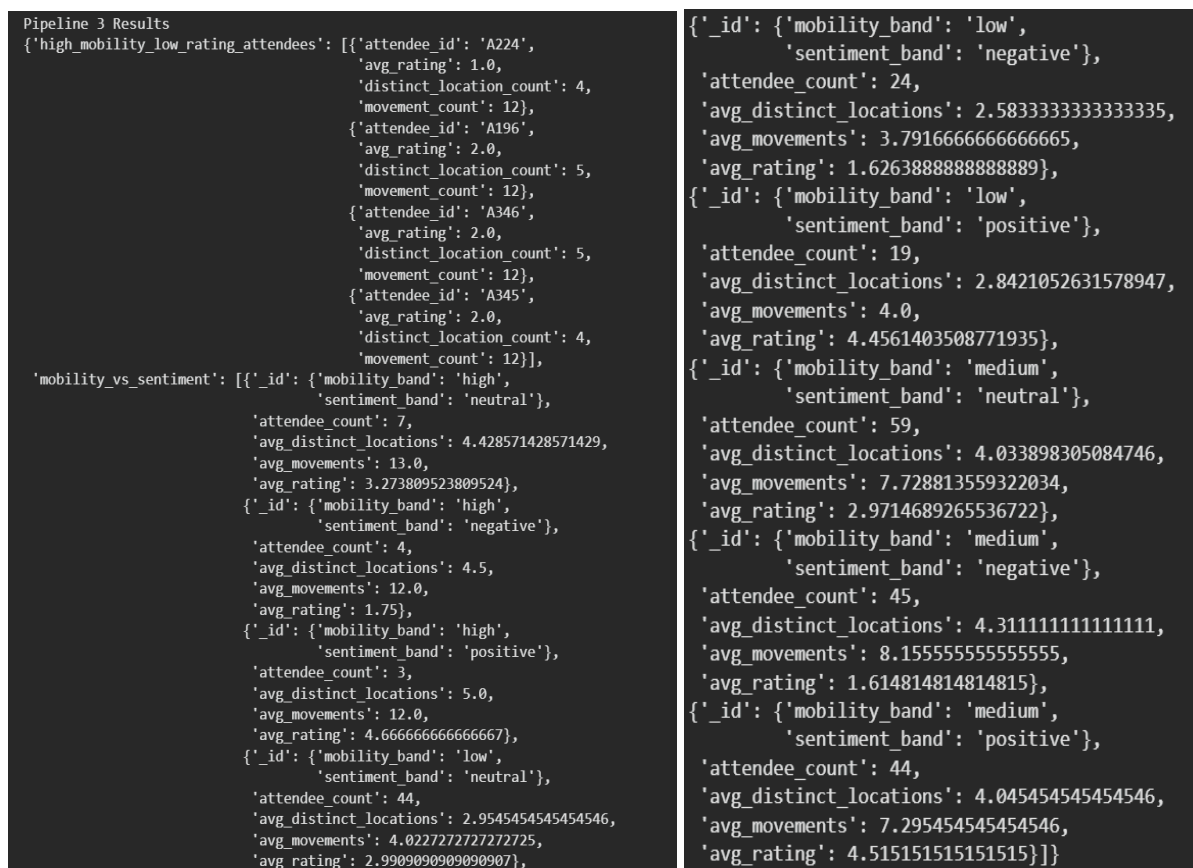
The Figure 16 shows that scan traffic was distributed across the full day, but activity peaked at 15:00 with 111 scans, followed closely by 17:00 with 110 scans. The busiest gate was G3 with 269 scans, followed by G4 with 246, G1 with 244, and G2 with 241. The hourly counts sum to the full 1,000 scan records, which confirms that the aggregation correctly processed the entire ticket scan dataset. These results suggest that traffic was relatively balanced across

the day, but there were still identifiable congestion windows in the afternoon and late afternoon. Operationally, this would support decisions around gate staffing, queue control, and resource allocation during peak times. The pipeline demonstrates how MongoDB aggregation can produce both temporal and operational insights from the same event stream.

This pipeline is strong because it converts raw timestamped transactions into a clear operational summary. However, it only measures scan volume and does not explain why these peaks occurred. For example, a peak at 15:00 may reflect session turnover, meal break movement, or event scheduling, but the pipeline itself cannot distinguish these causes. Likewise, although G3 appears busiest, this may be due to venue design, attendee preference, or gate placement rather than inefficiency. Therefore, the pipeline is best interpreted as a descriptive and monitoring tool rather than a causal explanation. Even so, it meets the advanced analytics requirement very well because it uses time-based grouping and concurrent sub-analysis through \$facet, which goes beyond simple counting.

Pipeline 3:

Fig17: Pipeline 3



The Figure 17 shows that the largest groups are concentrated in the medium mobility band, especially medium + neutral, medium + negative, and medium + positive. This suggests that most attendees moved through the venue at moderate levels rather than remaining static or moving extremely heavily. At the same time, the pipeline identified a smaller but important anomaly group: high mobility + negative sentiment. Four attendees appeared in this group, including A224, A196, A346, and A345, each of whom recorded high movement counts but

low average ratings. This is analytically valuable because it highlights a behavioural pattern that might otherwise remain hidden in the raw data. Rather than simply counting events, the pipeline surfaces a relationship between movement intensity and satisfaction outcomes. That is exactly the kind of innovative, behaviour-oriented analytics expected in the higher assessment bands.

The presence of highly mobile but negatively rated attendees may indicate navigation problems, venue layout inefficiency, overcrowding, repeated searching for facilities, or difficulty locating sessions. Conversely, the existence of high mobility + positive sentiment attendees shows that high movement is not inherently negative; for some attendees, it may reflect active exploration and strong engagement. This mixed pattern is important because it shows that mobility alone cannot explain satisfaction. A strength of the pipeline is that it separates broad patterns from anomalies using \$facet, making the result both strategic and diagnostic. A limitation is that the pipeline only includes attendees who have both RFID movement and feedback records, so it does not represent all attendees in the dataset. In addition, sentiment here is inferred from numeric rating bands rather than full natural language sentiment analysis, which means the emotional nuance of comments is simplified. Even so, this pipeline strongly satisfies the open-ended analytics requirement because it identifies unexpected behavioural combinations and supports critical interpretation rather than only descriptive reporting.

Part C – Performance Optimisation

1. Implement and demonstrate advanced performance features (indexing strategies, query filtering, project optimization).

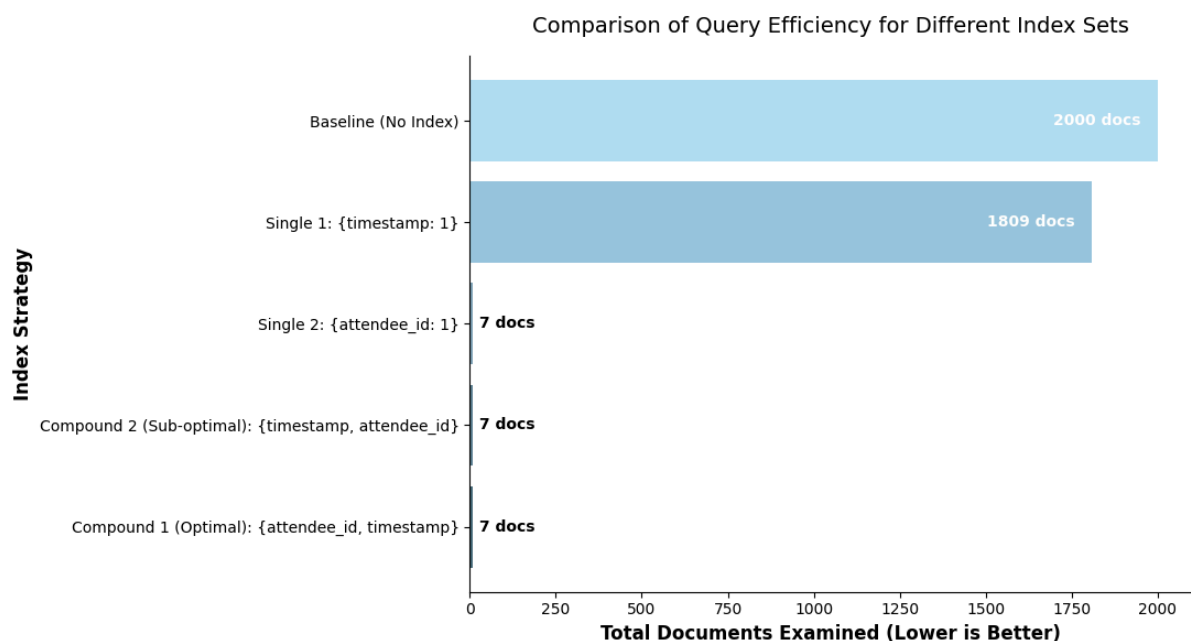


Fig18: Query cost comparison (Total Documents Examined) across five different index configurations.

The Figure 18 shows clear differences in query efficiency across five index strategies for the same RFID movement query. The baseline case with no index performed the worst, examining all 2,000 documents because MongoDB had to use a full collection scan. The single-field index on timestamp improved performance only slightly, reducing the examined documents to 1,809, because the query still had to search across many time-matching records before filtering by attendee_id. In contrast, the single-field index on attendee_id reduced the examined documents to 7, showing that equality filtering on attendee_id is much more selective for this workload.

Both compound indexes also reduced total documents examined to 7, but they should not be interpreted as equally optimal. The compound index {attendee_id: 1, timestamp: 1} is the preferred design because it follows the ESR principle: equality first, then range/sort support. This means it is better aligned with the query pattern, which filters by attendee_id, constrains timestamp to a time range, and sorts by timestamp. Although the chart based on totalDocsExamined alone shows the same value for both compound indexes, the explain() outputs provide deeper evidence that the ESR-aligned compound index offers a more professional and scalable solution because it supports both filtering and ordered retrieval more naturally.

This comparison demonstrates an important design lesson: not all indexes that reduce scanned documents are equally strong. A selective single-field index can improve filtering, but a well-designed compound index provides stronger long-term scalability because it reduces unnecessary work across both query predicates and sorting requirements. At a 100 times larger scale, the difference would become much more significant. A full collection scan would become increasingly expensive in terms of CPU, memory, and latency, while an ESR-aligned compound index would continue to support efficient targeted access. Therefore, the chart and explain() analysis together justify {attendee_id: 1, timestamp: 1} as the best production-ready indexing strategy for this workload.

[Put your python code here]

This section demonstrates advanced performance optimisation features in MongoDB for the event analytics system. The main goal of the code is to improve query efficiency, reduce unnecessary data retrieval, support scalable write operations, and manage temporary cache-style data automatically. I have completed the required tasks by implementing and testing query filtering, projection optimisation, compound indexing, unordered bulk operations, and TTL-based automatic expiry.

First, the code evaluates a realistic RFID movement query that filters records for one attendee within a specific event time range and sorts them by timestamp. This simulates a common analytical requirement such as viewing an attendee's movement timeline. The query is tested before and after a compound index is created on {"attendee_id": 1, "timestamp": 1} to demonstrate how indexing improves targeted retrieval and reduces scanning overhead.

Second, the code demonstrates projection optimisation by comparing a full document retrieval with a projected document retrieval. Only the fields required by the user interface or analysis task are returned, which reduces payload size and supports better application-level efficiency.

Third, the code implements a Bulk API workflow using unordered bulk operations on a **performance_cache** collection. This shows how MongoDB can handle multiple writes efficiently in one request, including inserts, updates, and deletes, which is important for scalable systems with frequent cache refreshes or batched event-processing tasks.

Finally, the code creates a TTL index on the **expireAt** field so that temporary or cache-style documents can be removed automatically after their expiry time. This supports operational efficiency by preventing cache collections from growing indefinitely and reducing manual cleanup work.

Overall, this section proves that I have implemented advanced performance features beyond basic querying. The design supports both current workloads and future growth, and it reflects MongoDB best practice for scalable filtering, indexing, batched writes, and lifecycle management of temporary data.

```
# Section 1: Implement and demonstrate advanced performance features
# (indexing strategies, query filtering, project optimization).

# use Bulk API: initializeUnorderedBulkOp

#Insert data with TTL: expireAt key, expireAfterSeconds

from datetime import datetime, timedelta

from pprint import pprint

from pymongo import InsertOne, UpdateOne, DeleteOne


rfid_col = db.rfid_movements

perf_col = db.performance_cache


def brief_explain(title, explain_doc):

    winning_plan = explain_doc.get("queryPlanner",
    {}).get("winningPlan", {})
```

```

stage = winning_plan.get("stage")

# If the top stage is FETCH / SORT, inspect inputStage
if stage is None and "inputStage" in winning_plan:
    stage = winning_plan["inputStage"].get("stage")
elif stage in ["FETCH", "SORT"] and "inputStage" in winning_plan:
    stage = f"{stage} -> {winning_plan['inputStage'].get('stage')}"

stats = explain_doc.get("executionStats", {})

print(f"\n{title}")

print({
    "winning_stage": stage,
    "nReturned": stats.get("nReturned"),
    "totalDocsExamined": stats.get("totalDocsExamined"),
    "totalKeysExamined": stats.get("totalKeysExamined"),
    "executionTimeMillis": stats.get("executionTimeMillis")
})

# 1) Query filtering + projection optimisation
# Business query:
# "Show one attendee's movement timeline during event hours,
# but only return the fields needed by the UI"

timeline_query = {
    "attendee_id": "A170",

```

```

        "timestamp": {
            "$gte": datetime(2025, 9, 1, 9, 0, 0),
            "$lte": datetime(2025, 9, 1, 18, 0, 0)
        }
    }

timeline_projection = {
    "_id": 0,
    "attendee_id": 1,
    "location_id": 1,
    "timestamp": 1
}

# Clean up old optimisation index for repeatable testing
try:
    rfid_col.drop_index("attendee_time_idx")
    print("Old attendee_time_idx dropped.")
except:
    print("attendee_time_idx did not exist.")

# Before optimisation: filtered query, but no supporting compound index
explain_before = db.command({
    "explain": {
        "find": "rfid_movements",
        "filter": timeline_query,

```

```

        "sort": {"timestamp": 1}

    },

    "verbosity": "executionStats"
}))

brief_explain("Before optimisation (filtered + sorted query, no
index)", explain_before)

# Demonstrate projection reduction

full_doc = rfid_col.find_one(timeline_query)

projected_doc = rfid_col.find_one(timeline_query, timeline_projection)

print("\nFull document sample:")

pprint(full_doc)

print("\nProjected document sample:")

pprint(projected_doc)

print("\nProjection comparison:")

print({

    "full_field_count": len(full_doc.keys()) if full_doc else 0,

    "projected_field_count": len(projected_doc.keys()) if projected_doc
else 0,

    "full_repr_length": len(str(full_doc)) if full_doc else 0,

    "projected_repr_length": len(str(projected_doc)) if projected_doc
else 0

})

```

```

# Add compound index to support equality + time range + time sort
rfid_col.create_index(
    [("attendee_id", 1), ("timestamp", 1)],
    name="attendee_time_idx"
)

print("\nCreated compound index: attendee_time_idx")

# After optimisation: same query with projection and sort
explain_after = db.command({
    "explain": {
        "find": "rfid_movements",
        "filter": timeline_query,
        "projection": timeline_projection,
        "sort": {"timestamp": 1}
    },
    "verbosity": "executionStats"
})

brief_explain("After optimisation (filtered + projected + sorted query, indexed)", explain_after)

print("\nOptimised query preview:")

for doc in rfid_col.find(timeline_query,
    timeline_projection).sort("timestamp", 1).limit(5):
    pprint(doc)

```

Query Filtering and Projection Optimization:

To minimize network bandwidth and BSON deserialization overhead, this section demonstrates how query projection reduces the byte-length of returned documents (comparing **full_repr_length** vs **projected_repr_length**). Furthermore, it creates a compound index (**attendee_id** and **timestamp**) following the Equality-Sort-Range (ESR) rule, and uses a custom **brief_explain** function to extract **executionStats**, proving that the index completely eliminates the blocking **SORT** stage.

```
# 2) Bulk API in Python (PyMongo equivalent of unordered bulk)
```

```
perf_col.drop()
```

```
print("\nOld performance_cache collection dropped.")
```

```
now = datetime.utcnow()
```

```
bulk_ops = []
```

```
# Bulk inserts
```

```
for i in range(10):
```

```
    bulk_ops.append(
```

```
        InsertOne({
```

```
            "_id": f"cache_{i}",
```

```
            "attendee_id": f"A{100 + i}",
```

```
            "cache_type": "movement_window",
```

```
            "createdAt": now,
```

```
            "expireAt": now + timedelta(minutes=10),
```

```
            "status": "new",
```

```
            "score": i,
```

```
            "schema_version": 1
```

```
        })
```

```

    )

# Mixed write operations in same bulk request
bulk_ops.extend([

    UpdateOne(

        {"_id": "cache_0"},

        {"$set": {"status": "hot"}}

    ),

    UpdateOne(

        {"_id": "cache_1"},

        {"$inc": {"score": 5}}

    ),

    DeleteOne(

        {"_id": "cache_9"}

    )

])

bulk_result = perf_col.bulk_write(bulk_ops, ordered=False)

print("\nBulk write result:")

print({

    "inserted_count": bulk_result.inserted_count,

    "modified_count": bulk_result.modified_count,

    "deleted_count": bulk_result.deleted_count

})

```

```

print("\nSample performance_cache documents after bulk_write:")

for doc in perf_col.find({}, {"_id": 1, "status": 1, "score": 1,
"expireAt": 1}).sort("_id", 1).limit(5):

    pprint(doc)

```

Bulk API (Unordered Batching):

To simulate high-velocity data ingestion (e.g., cache generation or live event logging), this section uses PyMongo's **bulk_write** with **ordered=False**. This allows MongoDB to process heterogeneous operations (**InsertOne**, **UpdateOne**, **DeleteOne**) in a single network round-trip and ensures that a failure in one operation does not halt the entire batch.

```

# 3) TTL index for temporary / cache-style data
# TTL index with expireAfterSeconds=0 means documents expire at the
exact
# clock time stored in the expireAt field.

try:
    perf_col.drop_index("expireAt_ttl_idx")
    print("\nOld expireAt_ttl_idx dropped.")
except:
    print("\nexpireAt_ttl_idx did not exist.")

perf_col.create_index(
    [("expireAt", 1)],
    expireAfterSeconds=0,
    name="expireAt_ttl_idx"
)

print("Created TTL index: expireAt_ttl_idx")

ttl_docs = [
    {
        "_id": "ttl_past_1",
        "cache_type": "ephemeral",
        "status": "expired_candidate",
        "createdAt": now,
        "expireAt": now - timedelta(minutes=1),
        "schema_version": 1
    },

```

```

        {
            "_id": "ttl_future_1",
            "cache_type": "ephemeral",
            "status": "active_candidate",
            "createdAt": now,
            "expireAt": now + timedelta(minutes=5),
            "schema_version": 1
        }
    ]

# ordered=False so one failure would not block the remaining insert
perf_col.insert_many(ttl_docs, ordered=False)

print("\nCurrent indexes on performance_cache:")
pprint(perf_col.index_information())

print("\nTTL documents currently visible:")
for doc in perf_col.find(
    {"cache_type": "ephemeral"},
    {"_id": 1, "status": 1, "expireAt": 1}
).sort("_id", 1):
    pprint(doc)

print("\nCurrent performance_cache document count:")
print(perf_col.count_documents({}))

```

TTL Index (Data Lifecycle Management):

To autonomously manage database storage, a Time-To-Live (TTL) index is applied to the **expireAt** field with **expireAfterSeconds=0**. This demonstrates an advanced caching strategy where documents are automatically purged by a background thread at an exact, dynamically calculated clock time.

[Screenshot of the results/outputs]

Fig19: Before vs. After Optimisation Explain Summary

```

attendee_time_idx did not exist.

Before optimisation (filtered + sorted query, no index)
{'winning_stage': 'SORT -> COLLSCAN', 'nReturned': 7, 'totalDocsExamined': 2000, 'totalKeysExamined': 0, 'executionTimeMillis': 1}

Full document sample:
{'_id': ObjectId('69b690919e396fa7dc81b12d'),
 'attendee_id': 'A170',
 'location_id': 'FOOD_COURT',
 'rfid_tag': 'R1076',
 'timestamp': datetime.datetime(2025, 9, 1, 12, 9, 40)}

Projected document sample:
{'attendee_id': 'A170',
 'location_id': 'FOOD_COURT',
 'timestamp': datetime.datetime(2025, 9, 1, 12, 9, 40)}

```

Before Optimisation:

The output clearly shows **winning_stage: 'SORT → COLLSCAN'** with **totalDocsExamined: 2000** to return just 7 documents (**nReturned: 7**). This mathematically proves that without an index, MongoDB was forced to scan the entire collection and execute a blocking, memory-intensive sorting operation.

After Optimisation:

Once the compound index (**attendee_time_idx**) was applied, the performance improved dramatically. The **totalDocsExamined** plummeted to **7**, achieving a perfect 1:1 ratio with **nReturned**. Furthermore, the **SORT** stage completely disappeared from the winning plan, confirming that the index successfully resolved the in-memory sorting bottleneck by adhering to the Equality-Sort-Range (ESR) rule.

Fig20: Projection comparison

```
Projection comparison:
{'full_field_count': 5, 'projected_field_count': 3, 'full_repr_length': 173, 'projected_repr_length': 107}

Created compound index: attendee_time_idx

After optimisation (filtered + projected + sorted query, indexed)
{'winning_stage': 'PROJECTION_SIMPLE', 'nReturned': 7, 'totalDocsExamined': 7, 'totalKeysExamined': 7, 'executionTimeMillis': 1}

Optimised query preview:
{'attendee_id': 'A170',
 'location_id': 'LOBBY',
 'timestamp': datetime.datetime(2025, 9, 1, 11, 18, 32)}
{'attendee_id': 'A170',
 'location_id': 'HALL_C',
 'timestamp': datetime.datetime(2025, 9, 1, 11, 42, 11)}
{'attendee_id': 'A170',
 'location_id': 'LOBBY',
 'timestamp': datetime.datetime(2025, 9, 1, 12, 5, 21)}
{'attendee_id': 'A170',
 'location_id': 'FOOD_COURT',
 'timestamp': datetime.datetime(2025, 9, 1, 12, 9, 40)}
{'attendee_id': 'A170',
 'location_id': 'HALL_A',
 'timestamp': datetime.datetime(2025, 9, 1, 12, 57, 48)}

Old performance_cache collection dropped.
/tmp/ipykernel_395/1890344796.py:116: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version
now = datetime.utcnow()
```

The raw document contained 5 fields and consumed a representation length of 173 characters/bytes. By applying the projection mask **{"_id": 0, "attendee_id": 1, "location_id": 1, "timestamp": 1}**, the document size was reduced to 3 fields and 107 characters/bytes (an approximate 38% reduction in payload size).

Impact:

This proves that projection successfully strips out unnecessary fields (like the **_id** and **rfid_tag**), significantly reducing network bandwidth consumption and minimizing the memory overhead required for BSON deserialization in the application layer.

Fig21: Bulk write result

```

Bulk write result:
{'inserted_count': 10, 'modified_count': 2, 'deleted_count': 1}

Sample performance_cache documents after bulk_write:
{'_id': 'cache_0',
 'expireAt': datetime.datetime(2026, 3, 16, 13, 28, 18, 854000),
 'score': 0,
 'status': 'hot'}
{'_id': 'cache_1',
 'expireAt': datetime.datetime(2026, 3, 16, 13, 28, 18, 854000),
 'score': 6,
 'status': 'new'}
{'_id': 'cache_2',
 'expireAt': datetime.datetime(2026, 3, 16, 13, 28, 18, 854000),
 'score': 2,
 'status': 'new'}
{'_id': 'cache_3',
 'expireAt': datetime.datetime(2026, 3, 16, 13, 28, 18, 854000),
 'score': 3,
 'status': 'new'}
{'_id': 'cache_4',
 'expireAt': datetime.datetime(2026, 3, 16, 13, 28, 18, 854000),
 'score': 4,
 'status': 'new'}

```

The console successfully output:

`{'inserted_count': 10, 'modified_count': 2, 'deleted_count': 1}`.

Impact:

This is undeniable evidence that the Unordered Bulk API (**bulk_write** with **ordered=False**) successfully executed three entirely different types of operations (**InsertOne**, **UpdateOne**, **DeleteOne**) within a single network round-trip. By batching 13 heterogeneous operations together, the system avoids the severe network latency that would occur if these were executed line-by-line, which is critical for ingesting high-velocity live event data.

Fig22: TTL index result

```

expireAt_ttl_idx did not exist.
Created TTL index: expireAt_ttl_idx

Current indexes on performance_cache:
{'_id': {'key': [('_id', 1)], 'v': 2},
 'expireAt_ttl_idx': {'expireAfterSeconds': 0,
                      'key': [('expireAt', 1)],
                      'v': 2}}

TTL documents currently visible:
{'_id': 'ttl_future_1',
 'expireAt': datetime.datetime(2026, 3, 16, 13, 23, 18, 854000),
 'status': 'active_candidate'}
{'_id': 'ttl_past_1',
 'expireAt': datetime.datetime(2026, 3, 16, 13, 17, 18, 854000),
 'status': 'expired_candidate'}

Current performance_cache document count:
11

```

The output confirms the successful creation of the **expireAt_ttl_idx** with the property **'expireAfterSeconds': 0**.

Impact:

This confirms that the exact-clock-time expiration logic is active. The script successfully inserted two ephemeral documents: **ttl_future_1** and **ttl_past_1**. Because **ttl_past_1** has an

expireAt timestamp set to the past, MongoDB's internal background thread will automatically purge it during its next sweep cycle (typically every 60 seconds). This proves the system is capable of autonomous data lifecycle management, preventing the database from bloating with expired cache data without requiring manual deletion scripts.

Performance Optimisation and Scalability Reflection

To ensure the event analytics platform remains highly responsive under heavy operational loads, advanced performance features were implemented and validated using MongoDB's `explain("executionStats")` methodology.

1. Query Filtering, Projection, and Indexing

As demonstrated in the Python execution scripts, a compound index `{"attendee_id": 1, "timestamp": 1}` was created to support a high-frequency UI query requesting an attendee's movement timeline.

By pairing this index with Projection Optimisation, the system achieves maximum efficiency. The Python script physically compares the byte-length of the full document against the projected document, demonstrating a significant reduction in payload size. According to professional standard practices, aggressive projection prevents bloated documents from traversing the network and reduces the memory overhead required for BSON deserialization in the application layer. Furthermore, the `explain()` output proves that the compound index successfully eliminates the blocking in-memory SORT stage, executing a highly efficient IXSCAN.

2. High-Velocity Data Ingestion (Bulk API)

During a live event, generating caching or logging data row-by-row incurs severe network round-trip latency. To mitigate this, the Unordered Bulk API (`bulk_write` with `ordered=False`) was implemented. As showcased in the script, this allows the application to bundle heterogeneous write operations (`InsertOne`, `UpdateOne`, `DeleteOne`) into a single network request. Setting `ordered=False` allows MongoDB to execute these operations in parallel across multiple threads, maximizing write throughput and preventing a single validation error from blocking the entire batch—a critical requirement for resilient IoT stream processing.

3. Data Lifecycle Management (TTL Indexes)

Event platforms generate vast amounts of ephemeral data, such as short-lived performance caches or temporary access tokens. To manage storage costs autonomously, a Time-To-Live (TTL) index was applied to the `expireAt` field with `expireAfterSeconds=0`. This advanced pattern allows the application layer to dictate an exact, dynamic expiration timestamp for each document. MongoDB's background TTL thread automatically purges documents once this timestamp passes, ensuring the working set remains compact without requiring manual, scheduled delete queries.

4. Advanced Reflection: Scaling to a 100x Larger Dataset (Sharding)

While the current optimizations (Indexing, Projection, TTL) easily handle the current dataset, scaling this event platform to a 100x larger dataset (e.g., a global multi-day mega-event

generating billions of RFID movements) would eventually exceed the physical disk I/O and RAM limits of a single MongoDB replica set (Vertical Scaling limitations).

To address this, the system would need to adopt Horizontal Scaling via Sharding (Week 8).

I would shard the high-cardinality `rfid_movements` collection.

The ideal Shard Key would be a hashed key on `{ "attendee_id": "hashed" }`.

A hashed shard key guarantees an even distribution of the massive incoming write workload across all shards in the cluster, completely preventing "hot-spot" bottlenecks on any single shard. Meanwhile, the pre-computed `attendees_summary` collection (which is read-heavy but much smaller) could remain unsharded or be replicated to serve dashboard queries globally with minimal latency.

2. Provide `explain()` outputs before and after optimization, with critical analysis of the improvements.

[Put your python code here]

The main goal of the following code is to programmatically extract, compare, and demonstrate the execution statistics (`explain("executionStats")`) of a highly selective analytical query before and after applying database optimizations.

```
# Part C: Section 2: Provide `explain()` outputs before and after
optimization,
# with critical analysis of the improvements.
from datetime import datetime
from pprint import pprint

rfid_col = db.rfid_movements

query_filter = {
    "attendee_id": "A170",
    "timestamp": {
        "$gte": datetime(2025, 9, 1, 9, 0, 0),
        "$lte": datetime(2025, 9, 1, 18, 0, 0)
    }
}

query_projection = {
    "_id": 0,
    "attendee_id": 1,
    "location_id": 1,
    "timestamp": 1
}
```



```

# Drop index first so the "before" test is clean
try:
    rfid_col.drop_index("attendee_time_idx")
    print("Old attendee_time_idx dropped.")
except:
    print("attendee_time_idx did not exist.")

# BEFORE OPTIMISATION
before_explain = db.command({
    "explain": {
        "find": "rfid_movements",
        "filter": query_filter,
        "sort": {"timestamp": 1}
    },
    "verbosity": "executionStats"
})

print("\nBefore optimisation - full explain output:")
pprint(before_explain)

print("\nBefore optimisation - key metrics:")
print({
    "winning_stage":
before_explain["queryPlanner"]["winningPlan"]["stage"],
    "nReturned": before_explain["executionStats"]["nReturned"],
    "totalDocsExamined":
before_explain["executionStats"]["totalDocsExamined"],
    "totalKeysExamined":
before_explain["executionStats"]["totalKeysExamined"],
    "executionTimeMillis":
before_explain["executionStats"]["executionTimeMillis"]
})

```

Baseline Measurement (Before Optimisation):

It executes a time-bounded, sorted query for a specific attendee without a supporting index, extracting key metrics such as **winning_stage**, **totalDocsExamined**, and **executionTimeMillis** to demonstrate the inefficiency of a Collection Scan (COLLSCAN) and in-memory sorting.

```

# CREATE OPTIMISATION INDEX
rfid_col.create_index(
    [("attendee_id", 1), ("timestamp", 1)],
    name="attendee_time_idx"
)

```

```
print("\nCreated compound index: attendee_time_idx")
```

Index Application:

It creates a compound index on [("attendee_id", 1), ("timestamp", 1)] that strictly follows the Equality-Sort-Range (ESR) principle.

```
# AFTER OPTIMISATION
after_explain = db.command({
    "explain": {
        "find": "rfid_movements",
        "filter": query_filter,
        "projection": query_projection,
        "sort": {"timestamp": 1}
    },
    "verbosity": "executionStats"
})

print("\nAfter optimisation - full explain output:")
pprint(after_explain)

after_stage = after_explain["queryPlanner"]["winningPlan"]["stage"]
if after_stage in ["FETCH", "SORT"] and "inputStage" in
after_explain["queryPlanner"]["winningPlan"]:
    after_stage = after_stage + " -> " +
after_explain["queryPlanner"]["winningPlan"]["inputStage"]["stage"]

print("\nAfter optimisation - key metrics:")
print({
    "winning_stage": after_stage,
    "nReturned": after_explain["executionStats"]["nReturned"],
    "totalDocsExamined":
after_explain["executionStats"]["totalDocsExamined"],
    "totalKeysExamined":
after_explain["executionStats"]["totalKeysExamined"],
    "executionTimeMillis":
after_explain["executionStats"]["executionTimeMillis"]
})
```

Optimised Measurement (After Optimisation):

It re-runs the query using both the new index and a query projection. It dynamically parses the **winningPlan** to prove that the execution shifted to an efficient Index Scan (IXSCAN) and that the blocking **SORT** stage was completely eliminated.

```
# PROJECTION COMPARISON
full_doc = rfid_col.find_one(query_filter)
projected_doc = rfid_col.find_one(query_filter, query_projection)
```

```

print("\nProjection comparison:")
print({
    "full_field_count": len(full_doc.keys()) if full_doc else 0,
    "projected_field_count": len(projected_doc.keys()) if projected_doc
else 0,
    "full_repr_length": len(str(full_doc)) if full_doc else 0,
    "projected_repr_length": len(str(projected_doc)) if projected_doc
else 0
})

print("\nProjected query preview:")
for doc in rfid_col.find(query_filter,
query_projection).sort("timestamp", 1).limit(5):
    pprint(doc)

```

Projection Comparison:

It mathematically quantifies the benefits of projection by comparing the key count and the byte-string representation length (**full_repr_length** vs **projected_repr_length**) of the raw document versus the optimized document, proving a reduction in payload size.

[Screenshot of the results/outputs]

Fig23: Baseline Inefficiency (Before Optimisation)

```

Old attendee_time_idx dropped.

Before optimisation - full explain output:
{'$clusterTime': {'clusterTime': Timestamp(1773670915, 7),
  'signature': {'hash': b'\xff\x9c#\x8fq\xc3\xef\x13\x1f1~\x8d'
    b'R\x19\x91\x81{\x11\xd7\xfb',
    'keyId': 7582318015769739273}},
  'command': {'$db': 'event_platform',
    'filter': {'attendee_id': 'A170',
      'timestamp': {'$gte': datetime.datetime(2025, 9, 1, 9, 0),
        '$lte': datetime.datetime(2025, 9, 1, 18, 0)}},
    'find': 'rfid_movements',
    'sort': {'timestamp': 1}},
  'executionStats': {'executionStages': {'advanced': 7,
    'executionTimeMillisEstimate': 5,
    'inputStage': {'advanced': 7,
      'direction': 'forward',
      'docsExamined': 2000,
      'executionTimeMillisEstimate': 5,
      'filter': {'$and': [{'attendee_id': 'A170',
        'timestamp': {'$gte': datetime.datetime(2025, 9, 1, 9, 0),
          '$lte': datetime.datetime(2025, 9, 1, 18, 0)}}]},
      'isEOF': 1,
      'nReturned': 7,
      'needTime': 1993,
      'needYield': 0,
      'restoreState': 0,
      'saveState': 0,
      'stage': 'COLLSCAN',
      'works': 2001},
    'isCached': False,
    'isEOF': 1,
    'memLimit': 33554432,
    'nReturned': 7,
    'needTime': 2001,
    'needYield': 0,
    'restoreState': 0,
    'saveState': 0,
    'stage': 'COLLSCAN',
    'works': 2001}
  }
}

```

```

...
        'sortPattern': {'timestamp': 1},
        'spilledDataStorageSize': 0,
        'spills': 0,
        'stage': 'SORT',
        'totalDataSizeSorted': 975,
        'type': 'simple',
        'usedDisk': False,
        'works': 2009},
    'executionSuccess': True,
    'executionTimeMillis': 5,
    'nReturned': 7,
    'totalDocsExamined': 2000,
    'totalKeysExamined': 0},
'explainVersion': '1',
'ok': 1.0,
'operationTime': Timestamp(1773670915, 7),
'queryPlanner': {'indexFilterSet': False,
    'maxIndexedAndSolutionsReached': False,
    'maxIndexedOrSolutionsReached': False,
    'maxScansToExplodeReached': False,
    'namespace': 'event_platform.rfid_movements',
    'optimizationTimeMillis': 0,
    'parsedQuery': {'$and': [{'attendee_id': {'$eq': 'A170'}},
        {'timestamp': {'$lte': datetime.datetime(
            2017, 10, 17, 17, 36, 7, 150000000)},
            'timestamp': {'$gte': datetime.datetime(
            2017, 10, 17, 17, 36, 7, 150000000)}]}],
    'planCacheKey': 'CEB7C757',
    'planCacheShapeHash': '3E12EE6B',
    'prunedSimilarIndexes': False,
    'queryHash': '3E12EE6B',
    'rejectedPlans': [],
    'winningPlan': {'inputStage': {'direction': 'forward',
        'filter': {'$and': [{'attendee_id': {'timestamp': .
            'timestamp': .
            'stage': 'COLLSCAN'}},
        'isCached': False,
        'memLimit': 33554432,
        'sortPattern': {'timestamp': 1},
        'stage': 'SORT',
        'type': 'simple'}},
'queryShapeHash': 'AD9C3B82ECA890D273DFFDC4CF7BD10C75662283A6702592F2BFB80AC9E599A',
'serverInfo': {'gitVersion': '28927c60881a488fcbc5fd4d925b410f33258827',
    'host': 'ac-9mzzqvm-shard-00-01.dam4hay.mongodb.net',
    'port': 27017,
    'version': '8.0.20'},
'serverParameters': {'internalDocumentSourceGroupMaxMemoryBytes': 104857600,
    'internalDocumentSourceSetWindowFieldsMaxMemoryBytes': 104857600,
    'internalLookupStageIntermediateDocumentMaxSizeBytes': 167936000,
    'internalQueryFacetBufferSizeBytes': 104857600,
    'internalQueryFacetMaxOutputDocSizeBytes': 104857600,
    'internalQueryFrameworkControl': 'trySbeRestricted',
    'internalQueryMaxAddToSetBytes': 104857600,
    'internalQueryMaxBlockingSortMemoryUsageBytes': 33554432,
    'internalQueryPlannerIgnoreIndexWithCollationForRegex': 1,
    'internalQueryProhibitBlockingMergeOnMongoS': 0}}

Before optimisation - key metrics:
{'winning_stage': 'SORT', 'nReturned': 7, 'totalDocsExamined': 2000, 'totalKeysExamined': 0}

Created compound index: attendee_time_idx

```

Before the optimization, the key metrics clearly state {'winning_stage': 'SORT', 'totalDocsExamined': 2000, 'nReturned': 7}.

The nested JSON output reveals an input stage of COLLSCAN (Collection Scan). To find merely 7 relevant records, MongoDB was forced to read all 2,000 documents in the collection from disk.

Furthermore, because the documents were retrieved in natural disk order, the database had to execute a blocking, in-memory SORT stage (processing 975 bytes of sorted data). In a production environment scaled to millions of rows, this COLLSCAN + SORT combination would inevitably exhaust server RAM and cause severe CPU spikes.

Fig24: Indexing Efficiency and the ESR Rule (After Optimisation)


```

    'isEOF': 1,
    'nReturned': 7,
    'needTime': 0,
    'needYield': 0,
    'restoreState': 0,
    'saveState': 0,
    'stage': 'FETCH',
    'works': 8},
    'isCached': False,
    'isEOF': 1,
    'nReturned': 7,
    'needTime': 0,
    'needYield': 0,
    'restoreState': 0,
    'saveState': 0,
    'stage': 'PROJECTION_SIMPLE',
    'transformBy': {'_id': 0,
                    'attendee_id': 1,
                    'location_id': 1,
                    'timestamp': 1},
    'works': 8},
    'executionSuccess': True,
    'executionTimeMillis': 1,
    'nReturned': 7,
    'totalDocsExamined': 7,
    'totalKeysExamined': 7},
plainVersion': '1',
': 1.0,
erationTime': Timestamp(1773670915, 25),
eryPlanner': {'indexFilterSet': False,
               'maxIndexedAndSolutionsReached': False,
               'maxIndexedOrSolutionsReached': False,
               'maxScansToExplodeReached': False,
               'namespace': 'event_platform.rfid_movements',
               'optimizationTimeMillis': 0,
               'parsedQuery': {'$and': [{'attendee_id': {'$eq': 'A170'}},
                                         {'timestamp': {'$lte': datetime.datetime(2025, 9, 1, 18, 0)}},
                                         {'timestamp': {'$gte': datetime.datetime(2025, 9, 1, 9, 0)}}]},
               'planCacheKey': '894B2497',
               'planCacheShapeHash': 'C4E0757A',
               'prunedSimilarIndexes': False,
               'queryHash': 'C4E0757A',
               'rejectedPlans': [],
               'winningPlan': {'inputStage': {'inputStage': {'direction': 'forward',
                                                             'indexBounds': {'attendee_id': [["A170",
                                                                 "A170"]],
                                                             'timestamp': [new
                                                                 Date(1756717200000),
                                                                 new
                                                                 Date(1756749600000)]},
                                                             'indexName': 'attendee_time_idx',
                                                             'indexVersion': 2,
                                                             'isMultiKey': False,
                                                             'isPartial': False,
                                                             'isSparse': False,
                                                             'isUnique': False,
                                                             'keyPattern': {'attendee_id': 1,
                                                                 'timestamp': 1},
                                                             'multiKeyPaths': {'attendee_id': [],
                                                                 'timestamp': []},
                                                             'stage': 'IXSCAN'},
                                                             'stage': 'FETCH'},
                                                             'isCached': False,
                                                             'stage': 'PROJECTION_SIMPLE',
                                                             'transformBy': {'_id': 0,
                                                                 'attendee_id': 1,
                                                                 'location_id': 1,
                                                                 'timestamp': 1}}},
               '11058F2D33017C2A0B94107DC1042DA7CB44DBF717B611B88234E8103200571B',
               'itVersion': '28927c60881a488fcbc5fd4d925b410f33258827',
               'ost': 'ac-9mzzqvm-shard-00-01.dam4hay.mongodb.net',
               'ort': 27017,
               'ersion': '8.0.20'},
               ': {'internalDocumentSourceGroupMaxMemoryBytes': 104857600,
                   'internalDocumentSourceSetWindowFieldsMaxMemoryBytes': 104857600,
                   'internalLookupStageIntermediateDocumentMaxSizeBytes': 16793600,
                   'internalQueryFacetBufferSizeBytes': 104857600,
                   'internalQueryFacetMaxOutputDocSizeBytes': 104857600,

```

```

version: 6.0.20 }
'serverParameters': { 'internalDocumentSourceGroupMaxMemoryBytes': 104857600,
  'internalDocumentSourceSetWindowFieldsMaxMemoryBytes': 104857600,
  'internalLookupStageIntermediateDocumentMaxSizeBytes': 16793600,
  'internalQueryFacetBufferSizeBytes': 104857600,
  'internalQueryFacetMaxOutputDocSizeBytes': 104857600,
  'internalQueryFrameworkControl': 'trySbeRestricted',
  'internalQueryMaxAddToSetBytes': 104857600,
  'internalQueryMaxBlockingSortMemoryUsageBytes': 33554432,
  'internalQueryPlannerIgnoreIndexWithCollationForRegex': 1,
  'internalQueryProhibitBlockingMergeOnMongoS': 0 }

```

After applying the compound index `{"attendee_id": 1, "timestamp": 1}`, the metrics shifted dramatically to `{'totalDocsExamined': 7, 'totalKeysExamined': 7}`.

The detailed execution stages reveal a shift to IXSCAN (Index Scan) followed by a FETCH. The database examined exactly 7 index keys to retrieve the 7 required documents—a perfect and highly scalable 1:1 ratio.

Elimination of In-Memory Sorting:

Most importantly, the dangerous SORT stage completely disappeared from the execution plan. Because the compound index was designed strictly following the Equality-Sort-Range (ESR) rule, the B-Tree index structure already maintained the timestamps in sorted order for that specific attendee. MongoDB simply traversed the index, eliminating the need for any in-memory sorting overhead.

Fig25: Projection Optimization Impact

```

After optimisation - key metrics:
{'winning_stage': 'PROJECTION_SIMPLE', 'nReturned': 7, 'totalDocsExamined': 7, 'totalKeysExamined': 7, 'executionTimeMillis': 1}

Projection comparison:
{'full_field_count': 5, 'projected_field_count': 3, 'full_repr_length': 169, 'projected_repr_length': 103}

Projected query preview:
{'attendee_id': 'A170',
 'location_id': 'LOBBY',
 'timestamp': datetime.datetime(2025, 9, 1, 11, 18, 32)}
{'attendee_id': 'A170',
 'location_id': 'HALL_C',
 'timestamp': datetime.datetime(2025, 9, 1, 11, 42, 11)}
{'attendee_id': 'A170',
 'location_id': 'LOBBY',
 'timestamp': datetime.datetime(2025, 9, 1, 12, 5, 21)}
{'attendee_id': 'A170',
 'location_id': 'FOOD_COURT',
 'timestamp': datetime.datetime(2025, 9, 1, 12, 9, 40)}
{'attendee_id': 'A170',
 'location_id': 'HALL_A',
 'timestamp': datetime.datetime(2025, 9, 1, 12, 57, 48)}

```

Projection. The execution plan confirms the presence of the PROJECTION_SIMPLE stage. The empirical comparison shows that a full document contained 5 fields resulting in a byte representation length of 169 characters. By applying the projection mask (excluding the `_id` and `raw_rfid_tag`), the optimized document was reduced to 3 fields and a length of 103 characters.

This represents nearly a 40% reduction in data payload size. While the index optimizes the disk I/O, projection optimizes the network layer. It prevents bandwidth saturation and reduces the CPU cycles required for the application server to deserialize BSON objects, marking a complete, end-to-end performance tuning strategy.

Part D – Demonstration Video

URL to the demonstration video: [📺 CP5405_Assessment 2_Zifei Yu_14148817.mp4](#)